

Linguistic Variation in Subreddit Communities

Authors: Mounami Kayitha, Varun Sardana, Prithvi Rao, Gabor Szita, Srishti Thapar

Introduction

Late 2022 saw the release of ChatGPT, an LLM that became one of the first to introduce AI to a mainstream audience. Since then, public discourse around AI has exploded, especially in online communities. One such large forum is Reddit. Various subreddits developed to allow users to interact and debate on the topic of AI. However, these subreddits also developed distinct cultures around the same topic. To explore this variation, we explored the relationships with AI for four subreddits: r/MachineLearning, r/artificial, r/singularity, and r/Futurology. Despite sharing a common topic, these communities appear to attract meaningfully different audiences, leading to questions of whether the language and focus of each subreddit is unique. To that end, we sought to answer whether a machine learning model can accurately classify which subreddit a post belongs to based solely on its text, and what does classification performance tell us about how these communities differ linguistically? We compared two NLP representations: TF-IDF bag of words via three classifiers, and fine-tuned BERT transformer models. Accounting for class imbalance, both approaches exceeded the baseline, with BERT reaching 75–76% accuracy and macro F1 of 0.72–0.73. From these relatively high results, we find that despite all four subreddits discussing AI, they attract different posters using different vocabularies, concerns, and ways of framing the general topic at hand.

Methods

Data Collection and Cleaning

Data was collected from four, finalized subreddits with high relevance to AI using Reddit's public JSON API. After initial surveying and hitting ~250 post pagination cap per endpoint, multiple endpoints were queried to maximize dataset size: current year, all time top posts, currently trending (hot), and most controversial posts of the past year. Duplicates were removed via unique post ID and only self-text posts (text only, no links, no images) that had body text were kept. Our time period was capped from January 2023 to May 2026 (collection day), to ensure posts are after ChatGPT's releases.

Data Preprocessing

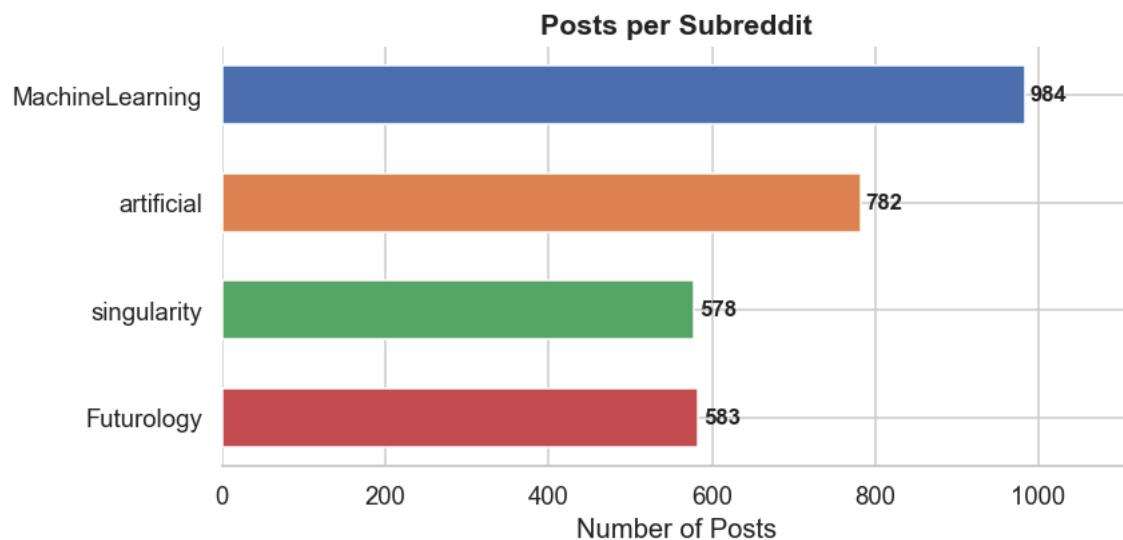
Title and body were combined before removing URLs, non-alphabetic characters, extra whitespace which further cleaned the data. Standard English stopwords were removed as well as an extended list of Reddit-specific words that only provided a source of noise. Each token was lemmatized such that they are reduced to their base using WordNetLemmatizer.

The final dataset contained 2972 posts across four subreddits:

- r/MachineLearning: 984 posts
- r/artificial: 782 posts
- r/Futurology: 583 posts
- r/singularity: 578 posts

Each of these observations represents a single Reddit post and includes information such as the subreddit, title, body text, score, upvote ratio, number of comments, date created, and combined text.

As shown in Figure 1, the distribution of posts in each subreddit class was not balanced. r/Machine Learning had roughly 1.7x more posts than r/singularity. As a result, a naive classifier that always predicted a post to be from Machine Learning would achieve 33.6% accuracy without any real training. Therefore, a macro F1 score was used as the primary metric for evaluating model strength by averaging F1 scores across all classes.



Posts Per Subreddit. r/Machine Learning (987) had roughly 1.7x more posts than r/singularity (578).

Exploratory Data Analysis

These four subreddits both differ and are similar in various other metrics. Figure 2 shows the similarities in post lengths across the communities, mostly around the 250 mark with

some outliers surpassing 2500.

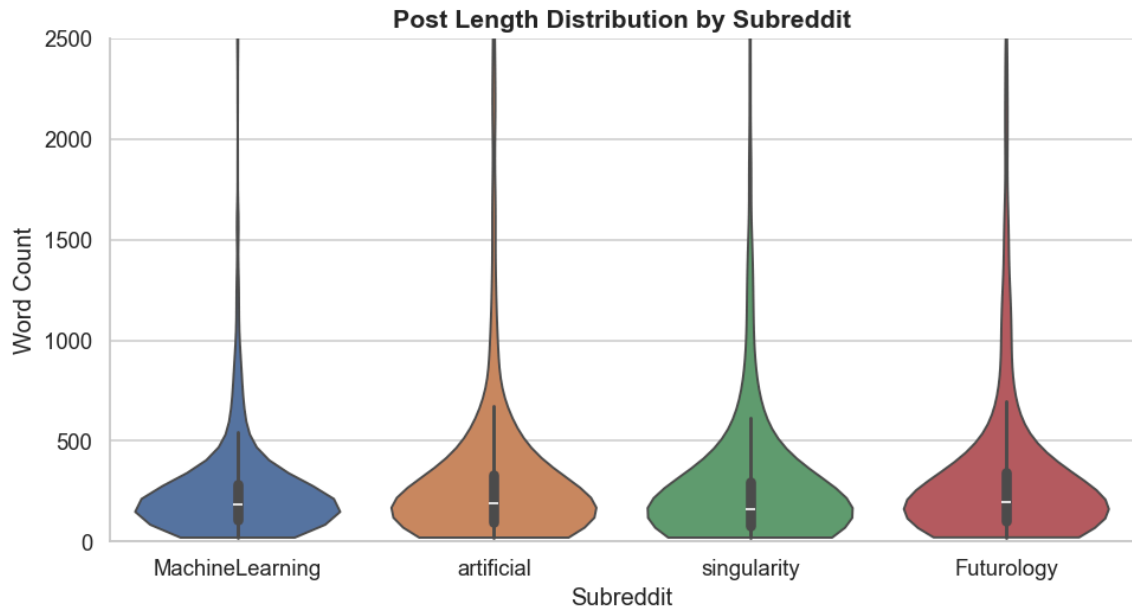
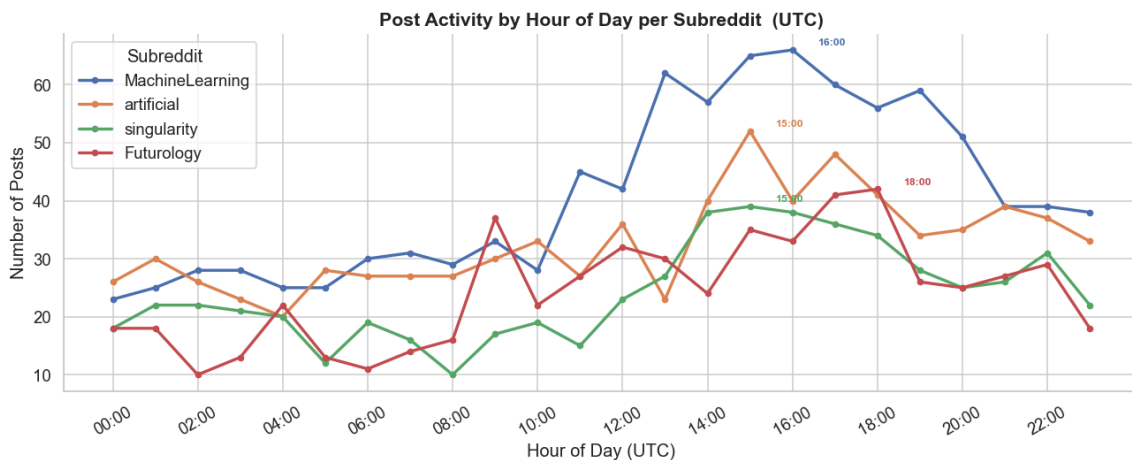


Figure 2: Length of posts does not vary drastically by community.

Interestingly, there is a high influx of posting activity across subreddits between 2:00PM UTC and 4:00PM UTC, with r/MachineLearning having a much higher post volume (Figure 3).

Figure 3:



TF-IDF Feature Extraction

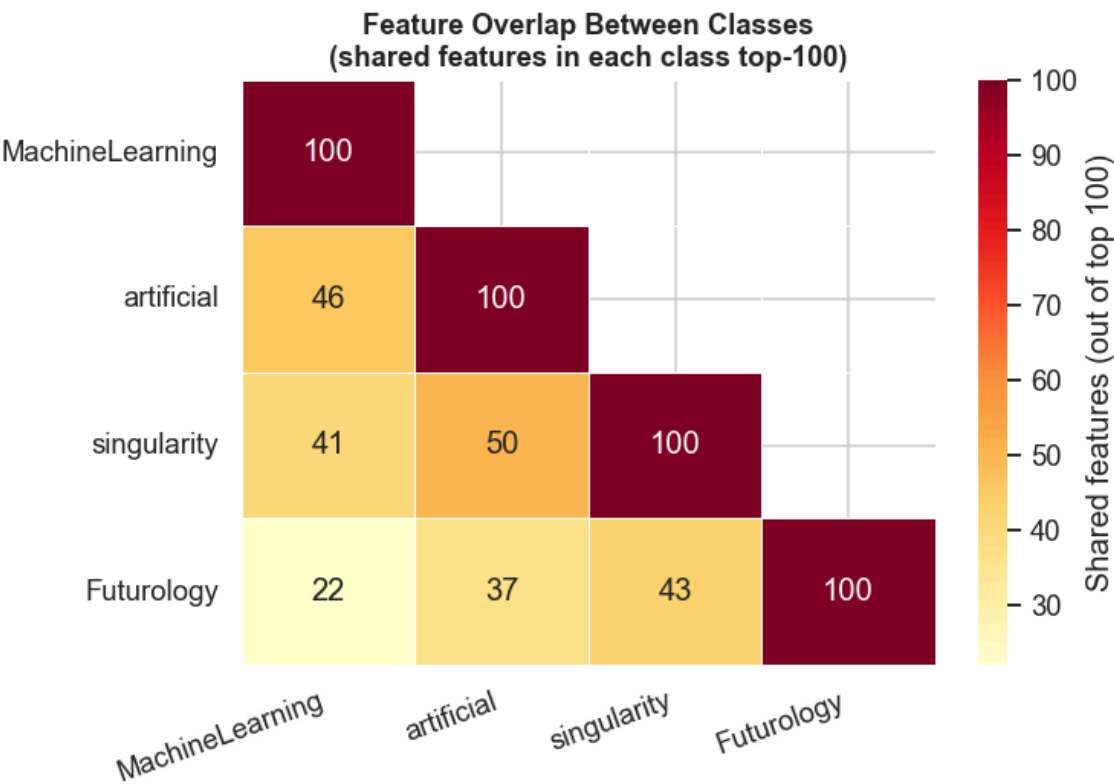
After cleaning, we then converted posts into numeric features using TF_IDF vectorization. This provided higher weightage to words and phrases that are important in a post but not common across the entire dataset. The vectorizer was set up with 15,000 max features (both unigram and bigram) and a minimum and maximum document

frequency to remove rare and common terms. This procuded a $2,927 \times 15,000$ sparse matrix that is 99.23% sparse.

The data was split into an 80/20 training-test with stratification by subreddit. This resulted in 2,341 posts for training and 586 posts for testing. Stratification was used due to the subreddit class uneven distribution.

Before training, we explored the vocabulary commonalities between subreddits (Figure 4).

Figure 4: The following TF-IDF matrix shows the feature overlap between classes.



We took the top 100 features per class and counted how many features each pair of subreddits share. The result is shown as a lower-triangle heatmap.

The diagonal is always 100 (a class shares all 100 of its own features with itself). The interesting numbers are the off-diagonal ones. A high number like 60 means two communities share 60% of their most characteristic features, meaning they are linguistically similar and the classifier will struggle to tell them apart. A low number like 20 means only 20% overlap (well-separated, easy to classify).

We can see in the confusion matrix that singularity and artificial share the most features, and Futurology and MachineLearning share the least features (most different audiences). Therefore, we expect that singularity and artificial to be the hardest for the models to tell apart, and Futurology and MachineLearning to be the easiest for the models to tell apart.

Models

Three classifiers were trained using the TF-IDF matrix, Multinomial Naive Bayes, Logistic Regression, and LinearSVC. These models train on word frequencies where Naive Bayes assumes feature independence, LogReg applies weight training (parameters: `c=2`, `class_weight = "balanced"`), and LinearSVC classifies via high-dimensional boundaries. These models were used as both baselines (Naive Bayes) and because some like LinearSVC are known for performing well in high-dimensional, sparse datasets.

To test the importance of contextual information, BERT based classifiers were fine-tuned (BERT and DistilBERT). BERT reads the full token "sentence" to produce representations of meaning with respect to context. This allows the model to integrate contextual cues for classification.

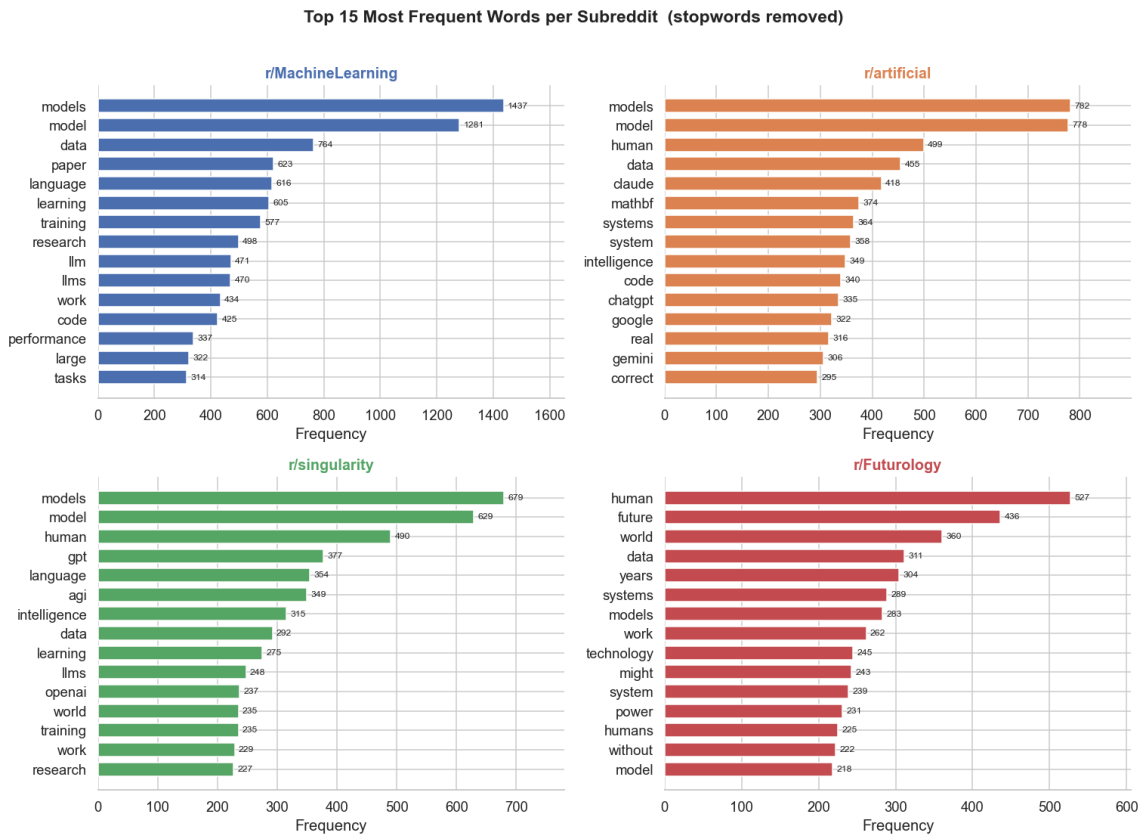
We also fine-tuned two BERT models to classify the reddit posts: `bert-base-uncased` and `distilbert-base-uncased`. These models are far more powerful than the simple TF-IDF-based models above.

Training both models allowed us to determine key characteristics of the subreddits including subtopical variation and linguistic differences.

Results

`r/MachineLearning` featured more academic terms like model, training, paper, and llm. `r/Futurology` leans toward more generic terms like human, future and world. `r/artificial` concentrated on LLM products like claude and chatgpt, and `r/singularity` uses vocab like agi, gpt. TF-IDF analysis confirmed that each subreddit has a somewhat unique vocabulary. Figure 7 describes the top 15 features per subreddit across training posts of each class and showcases the unique vocabulary we expected to see and would indicate high model classification macro F1. There is some overlap in vocabulary that is ranked high, like "model", which is a byproduct of the TF-IDF method used with a low document count.

Figure 5:



All three TF-IDF classifiers were evaluated to exceed both the random baseline (25%) and majority-class baseline (33.6%), confirming that each subreddit's postings have sufficient for distinguishing them. Looking at Figure 9a, r/singularity and r/artificial are commonly mixed with one another which is a trend across all models. r/Futurology improves with complexity and r/MachineLearning has consistently high F1 scores.

Figure 6a: Results for the three TF-IDF models:

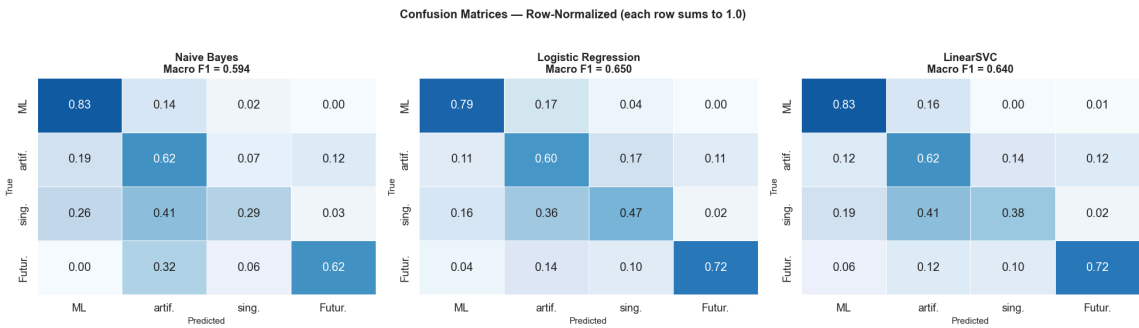
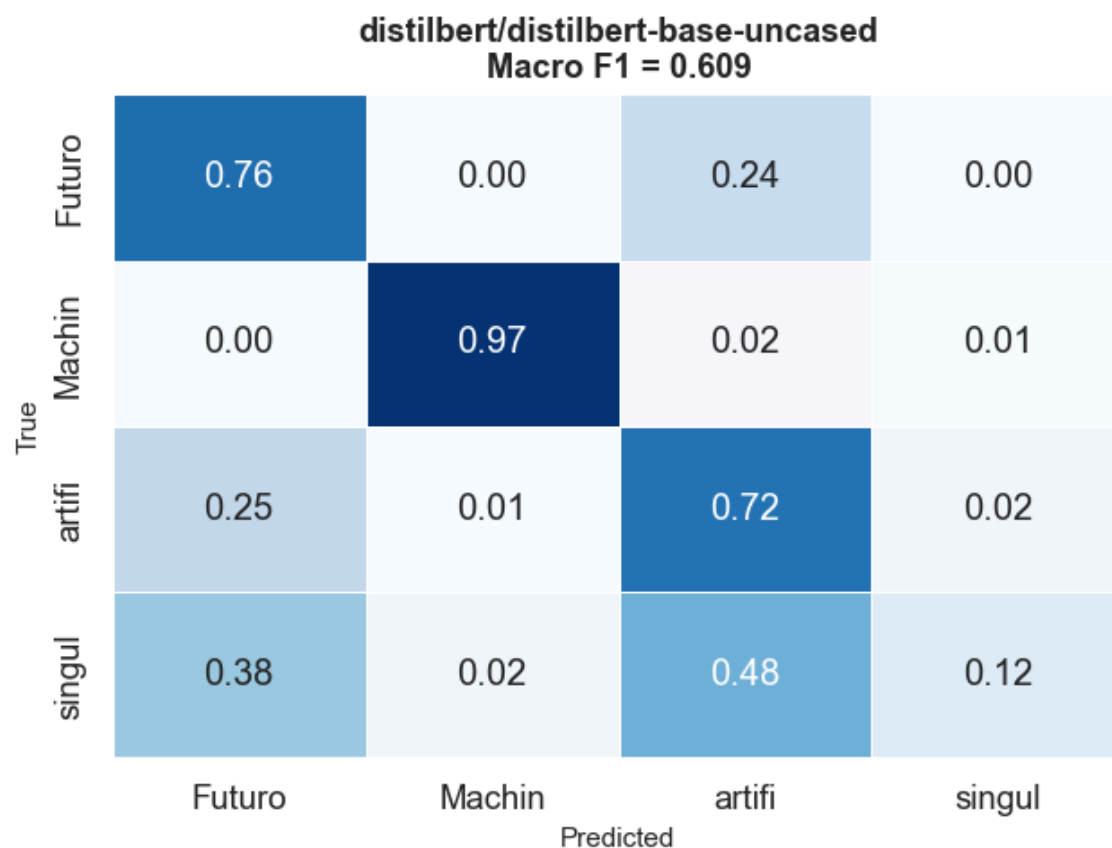


Figure 6b: Results for the BERT models:

Confusion Matrices — Row-Normalized (each row sums to 1.0)



Conclusion

We trained five models to classify Reddit posts across four AI subreddits: Naive Bayes (macro F1 = 0.594), Logistic Regression (F1 = 0.650), LinearSVC (67.2% accuracy, F1 = 0.640), and two fine-tuned BERT models (75-76% accuracy, F1 = 0.72-0.73). All models produced well above the 33.6% majority-class baseline. `distilbert-base-uncased` matched the accuracy of `bert-base-uncased` and even performed slightly better, despite being 40% smaller, making it the best practical choice for efficiency and accuracy.

Per-class results were consistent across all models: `r/MachineLearning` was easiest due to its distinctive vocabulary, and `r/singularity` was hardest because it overlaps linguistically with every other community. The `r/artificial`-`r/singularity` confusion was the most common error across all approaches, a pattern we predicted during the EDA before any of the models were trained as seen with the 50% vocabulary overlap.

A key limitation of this analysis is that we did not control for topic across subreddits. This could mean the classifier may be learning topical differences in some cases rather than purely linguistic ones, and we cannot separate the two without extremely limiting the

dataset. Further work could be to explore feature interpretability analysis. Identifying which words and phrases most drive classification per subreddit, via dimension reduction, could also us to separate whether semantics or topics are what separate these subreddits.

Overall, classification and exploration of these subreddits supports the idea that r/MachineLearning, r/singularity, r/artificial, r/Futurology, albeit related to the topic of AI, do exhibit linguistic differences that can be detected by both complex and classical NLP models.

Linguistic Variation in AI Subreddit Communities

Authors: Mounami Kayitha, Varun Sardana, Prithvi Rao, Gabor Szita, Srishthi Thapar

Course: PSTAT 134, Statistical Data Science

```
In [1]: import os
        from pathlib import Path

        # Set working directory to project root – runs once for the entire notebook
        _notebook_dir = Path().resolve()
        _project_root = _notebook_dir.parent if _notebook_dir.name == 'notebooks' else
        os.chdir(_project_root)
        print('Working directory:', os.getcwd())

        # Skip live Reddit API calls if the dataset is already collected
        _SKIP_COLLECTION = Path('data/reddit_ai_posts_CLEAN.csv').exists()
        print(f'Dataset already collected: {_SKIP_COLLECTION}')
```

Working directory: /Users/varunsardana/PSTAT-134-Final-Project

Dataset already collected: True

Section 1 — Data Collection

```
In [2]: import requests
        import pandas as pd
        import time
```

```
In [3]: headers = {"User-Agent": "pstat134project/1.0 (academic research)"}

        # The 4 subreddits we are collecting from
        subreddits = ["technology", "artificial", "singularity", "MachineLearning"]

        # Keep the class size balanced so each subreddit contributes about the same
        POSTS_PER_SUB = 200
```

```
In [4]: def get_posts(subreddit, limit=200):
        posts = []
        after = None

        while len(posts) < limit:
            url = f"https://www.reddit.com/r/{subreddit}/top.json?limit=100&t=ye
            if after:
                url += f"&after={after}"
```

```

response = requests.get(url, headers=headers)
data = response.json()

children = data["data"]["children"]
if not children:
    break

for post in children:
    p = post["data"]
    posts.append({
        "id": p["id"],
        "title": p["title"],
        "selftext": p["selftext"],
        "upvote_ratio": p["upvote_ratio"],
        "num_comments": p["num_comments"],
        "is_self": p["is_self"],
        "edited": p["edited"],
        "created_utc": p["created_utc"],
        "subreddit": p["subreddit"]
    })

after = data["data"]["after"]
if not after:
    break

time.sleep(1) # sleep 1 sec to be polite to Reddit's servers

return posts[:limit]

```

```

In [5]: if not _SKIP_COLLECTION:
        all_posts = []

        for sub in subreddits:
            print(f"Collecting from r/{sub}...")
            posts = get_posts(sub, POSTS_PER_SUB)
            all_posts.extend(posts)
            print(f"Got {len(posts)} posts")
            time.sleep(2)

        print(f"\nTotal posts collected: {len(all_posts)}")
    else:
        print('Skipped – data already collected')

```

Skipped – data already collected

```

In [6]: if not _SKIP_COLLECTION:
        df = pd.DataFrame(all_posts)
        # Preview it
        print(df.shape)
        print(df["subreddit"].value_counts())
        df.head()
    else:
        print('Skipped – data already collected')

```

Skipped – data already collected

```
In [7]: if not _SKIP_COLLECTION:
        # Save raw data to CSV
        df.to_csv("data/raw/reddit_raw.csv", index=False)
        print("Saved to data/raw/reddit_raw.csv")
    else:
        print('Skipped – data already collected')
```

Skipped – data already collected

```
In [8]: if not _SKIP_COLLECTION:
        missing_body = df[df["selftext"].isin(["", "[removed]", "[deleted]"])]
        print(f"Posts with missing/removed body text: {len(missing_body)}")
        print(f"\nMissing by subreddit:")
        print(missing_body["subreddit"].value_counts())
    else:
        print('Skipped – data already collected')
```

Skipped – data already collected

Data Collection Summary

- **API:** Reddit public JSON API
- **Subreddits we used:** r/MachineLearning, r/artificial, r/singularity, r/Futurology
- **Endpoints:** top/year, top/all, hot, controversial/year
- **Filter:** self-text posts ≥20 words, duplicates removed by post ID
- **Output File:** data/reddit_ai_posts_CLEAN.csv — 2,927 posts

Part 2 — Expanding From What We Know

The initial scrape hit Reddit's ~250-post endpoint due to which we were unable to collect more posts. To grow the dataset we go through three additional endpoints per subreddit, filter for posts with body text, check for duplication against already-collected IDs, and then add.

```
In [9]: import os
import requests
import pandas as pd
import numpy as np
import time
from datetime import datetime
from pathlib import Path

# Fix working directory – same pattern used in notebooks 02, 03, 04
notebook_dir = Path().resolve()
project_root = notebook_dir.parent if notebook_dir.name == 'notebooks' else
os.chdir(project_root)
print('Working directory:', os.getcwd())

SUBREDDITS = ["MachineLearning", "artificial", "singularity", "Futurology"]
HEADERS = {"User-Agent": "pstat134project/2.0 (academic research)"}
```

```

LIMIT_PER_REQ = 100
TARGET_PER_ENDPOINT = 250

NEW_ENDPOINTS = [
    ("top", "all"),
    ("hot", None),
    ("controversial", "year"),
]

BAD_SELFTEXT = {"", "[removed]", "[deleted]"}

def fetch_listing(subreddit, sort, timeframe=None, target=TARGET_PER_ENDPOINT):
    posts = []
    after = None
    while len(posts) < target:
        url = f"https://www.reddit.com/r/{subreddit}/{sort}.json?limit={LIMIT_PER_REQ}"
        if timeframe:
            url += f"&t={timeframe}"
        if after:
            url += f"&after={after}"
        try:
            resp = requests.get(url, headers=HEADERS, timeout=15)
            if resp.status_code == 429:
                print(f"Rate-limited - sleeping 30s...")
                time.sleep(30)
                continue
            if resp.status_code != 200:
                print(f"HTTP {resp.status_code} - skipping")
                break
            data = resp.json()
        except Exception as e:
            print(f"Error: {e} - skipping")
            break
        children = data.get("data", {}).get("children", [])
        if not children:
            break
        for child in children:
            posts.append(child["data"])
        after = data["data"].get("after")
        if not after or len(posts) >= target:
            break
        time.sleep(1.2)
    return posts[:target]

def build_row(p, subreddit):
    selftext = p.get("selftext", "") or ""
    title = p.get("title", "") or ""
    has_body = selftext.strip() not in BAD_SELFTEXT and len(selftext.strip()) > 0
    text = (title + " " + selftext).strip() if has_body else title
    created_utc = p.get("created_utc", 0)
    try:
        created_dt = datetime.utcfromtimestamp(created_utc).strftime("%Y-%m-%d %H:%M")
    except:
        created_dt = ""
    return {
        "id": p.get("id", ""),

```

```

        "subreddit":    subreddit,
        "title":        title,
        "selftext":     selftext,
        "text":         text,
        "text_word_count": len(text.split()),
        "has_body":     has_body,
        "upvote_ratio": p.get("upvote_ratio", np.nan),
        "num_comments": p.get("num_comments", 0),
        "score":        p.get("score", 0),
        "is_self":      p.get("is_self", False),
        "edited":       p.get("edited", False),
        "created_utc":  created_utc,
        "created_dt":   created_dt,
        "url":          p.get("url", ""),
        "permalink":    p.get("permalink", ""),
    }

```

```
print("Functions defined. Ready to collect.")
```

Working directory: /Users/varunsardana/PSTAT-134-Final-Project
 Functions defined. Ready to collect.

```

In [10]: if not _SKIP_COLLECTION:
import os
from pathlib import Path

notebook_dir = Path().resolve()
project_root = notebook_dir.parent if notebook_dir.name == 'notebooks' else
os.chdir(project_root)
print('Working directory:', os.getcwd())

existing_df = pd.read_csv("data/reddit_ai_posts_CLEAN.csv")
existing_ids = set(existing_df["id"].astype(str))
print(f"Existing dataset: {len(existing_df)} posts, {len(existing_ids)} e
print(f"Current counts per subreddit:")
print(existing_df["subreddit"].value_counts().to_string())
print()
new_rows = []

for sub in SUBREDDITS:
    sub_new = 0
    for sort, timeframe in NEW_ENDPOINTS:
        label = f"{sort}/{timeframe}" if timeframe else sort
        print(f"  r/{sub} - {label} ...", end=" ", flush=True)

        raw_posts = fetch_listing(sub, sort, timeframe)
        added = 0

        for p in raw_posts:
            pid = str(p.get("id", ""))
            if pid in existing_ids:
                continue

            row = build_row(p, sub)

            if not row["has_body"]:

```

```

        continue
    if row["text_word_count"] < 20:
        continue

    new_rows.append(row)
    existing_ids.add(pid)
    added += 1
    sub_new += 1

    print(f"{added} new posts with body text")
    time.sleep(2)

    print(f" → r/{sub} total new: {sub_new}")
    print()

    print(f"Total new posts collected: {len(new_rows)}")
else:
    print('Skipped – data already collected')

```

Skipped – data already collected

```

In [11]: if not _SKIP_COLLECTION:
    new_df = pd.DataFrame(new_rows)
    combined = pd.concat([existing_df, new_df], ignore_index=True)

    combined = combined.drop_duplicates(subset="id", keep="first")

    combined = combined.sort_values(["subreddit", "created_utc"]).reset_index()

    print("=== Combined Dataset ===")
    print(f"Total posts : {len(combined)}")
    print(f"New posts   : {len(new_df)}")
    print()
    print("Posts per subreddit (before → after):")
    before = existing_df["subreddit"].value_counts()
    after = combined["subreddit"].value_counts()
    for sub in ["MachineLearning", "artificial", "singularity", "Futurology":
        b = before.get(sub, 0)
        a = after.get(sub, 0)
        gain = a - b
        print(f" {sub:20s}: {b:4d} → {a:4d} (+{gain})")

    print("Word count stats on new posts:")
    print(new_df["text_word_count"].describe().round(1).to_string())
else:
    print('Skipped – data already collected')

```

Skipped – data already collected

```
In [12]: if not _SKIP_COLLECTION:
    output_path = Path("data/reddit_ai_posts_CLEAN.csv")
    combined.to_csv(output_path, index=False)
    print(f"Saved expanded dataset to {output_path}")
    print(f"Total posts: {len(combined)}")

    else:
        print('Skipped – data already collected')
```

Skipped – data already collected

```
In [13]: import pandas as pd
    _df_check = pd.read_csv('data/reddit_ai_posts_CLEAN.csv')
    print(f'Dataset loaded: {len(_df_check)} posts across {_df_check["subreddit"]}')
    print(_df_check['subreddit'].value_counts())
```

Dataset loaded: 2927 posts across 4 subreddits

subreddit

MachineLearning 984

artificial 782

Futurology 583

singularity 578

Name: count, dtype: int64

Section 2 — Data Cleaning & Exploratory Data Analysis

```
In [14]: import os
    from pathlib import Path

    notebook_dir = Path().resolve()
    project_root = notebook_dir.parent if notebook_dir.name == "notebooks" else
    os.chdir(project_root) # keep paths relative

    print("Working directory:", os.getcwd())
```

Working directory: /Users/varunsardana/PSTAT-134-Final-Project

```
In [15]: import pandas as pd
    import numpy as np
    import re
    import matplotlib.pyplot as plt
    import matplotlib.ticker as mticker
    import seaborn as sns
    import nltk
    from collections import Counter
    from nltk.corpus import stopwords
    from nltk.stem import WordNetLemmatizer
    from sklearn.feature_extraction.text import TfidfVectorizer
```

```
nlTK.download("stopwords", quiet=True)
nlTK.download("wordnet", quiet=True)
nlTK.download("punkt", quiet=True)

# consistent colors for each subreddit
PALETTE = {
    "MachineLearning": "#4C72B0",
    "artificial":      "#DD8452",
    "singularity":     "#55A868",
    "Futurology":      "#C44E52",
}
SUB_ORDER = ["MachineLearning", "artificial", "singularity", "Futurology"]

# shared plot style
sns.set_theme(style="whitegrid", font_scale=1.1) # default style
plt.rcParams["figure.dpi"] = 110
```

```
In [16]: df = pd.read_csv("data/reddit_ai_posts_CLEAN.csv", parse_dates=["created_dt"]

print(f"Dataset shape : {df.shape[0]} posts x {df.shape[1]} columns")
print(f>Date range    : {df['created_dt'].min().date()} → {df['created_dt'].max().date()}")
print()
print("Posts per subreddit:")
print(df["subreddit"].value_counts())
print()
print(df.head(3))
```


Dataset shape : 2927 posts x 16 columns

Date range : 2016-07-29 → 2026-05-26

Posts per subreddit:

subreddit

MachineLearning 984

artificial 782

Futurology 583

singularity 578

Name: count, dtype: int64

	id	subreddit	title \
0	101f9z4	Futurology	Pulling together different technologies to mak...
1	10ehigz	Futurology	Hello, I am Hassan Ragab. I am an interdiscipl...
2	10s52md	Futurology	I finally think the concept of AGI is misleadi...

	selftext \
0	Here are some fascinating articles and links a...
1	I'm an Egyptian-born U.S.-based architect and ...
2	TL;DR: (and it's long) What I am trying to arg...

	text	text_word_count \
0	Pulling together different technologies to mak...	204
1	Hello, I am Hassan Ragab. I am an interdiscipl...	143
2	I finally think the concept of AGI is misleadi...	1725

	has_body	upvote_ratio	num_comments	score	is_self	edited	created_utc
0	True	0.85	163	465	True	True	1.672673e+09
1	True	0.57	25	8	True	True	1.673976e+09
2	True	0.54	59	5	True	True	1.675382e+09

	created_dt	url \
0	2023-01-02 15:27:59	https://www.reddit.com/r/Futurology/comments/1...
1	2023-01-17 17:11:53	https://www.reddit.com/r/Futurology/comments/1...
2	2023-02-02 23:54:14	https://www.reddit.com/r/Futurology/comments/1...

	permalink
0	/r/Futurology/comments/101f9z4/pulling_togethe...
1	/r/Futurology/comments/10ehigz/hello_i_am_hass...
2	/r/Futurology/comments/10s52md/i_finally_think...

```
In [17]: df["text"] = df["text"].fillna("")
```

```
In [18]: def clean_text(text):
    text = text.lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"^[a-z\s]", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["full_text"] = df["text"].apply(clean_text)
```

```
In [ ]: stop_words = set(stopwords.words("english")) # removing the repeated and un
lemmatizer = WordNetLemmatizer() # getting the words to their base form here
```

```
def preprocess(text):
    tokens = text.split()
    tokens = [t for t in tokens if t not in stop_words]
    tokens = [lemmatizer.lemmatize(t) for t in tokens]
    return " ".join(tokens)

df["full_text"] = df["full_text"].apply(preprocess)
print("Sample preprocessed text:")
print(df["full_text"].iloc[0][:200])
```

Sample preprocessed text:

pulling together different technology make interstellar colonization possible
fascinating article link seemingly unrelated technology baby born year from
zen embryo seed thousand year old able sprout grow

We remove common English stopwords plus a custom list of Reddit words we personally want to remove ("like", "just", "going") that appear across all four subreddits but carry no value to us.

Then we perform Lemmatization which essentially maps words to their base form so "running" and "runs" count as the same token.

```
In [20]: before = len(df)
df = df[df["full_text"].apply(lambda x: len(x.split()) >= 3)]
after = len(df)
print(f"Rows before filter : {before}")
print(f"Rows after filter : {after} (dropped {before - after})")
print()
print("Posts per subreddit after cleaning:")
print(df["subreddit"].value_counts())
```

```
Rows before filter : 2927
Rows after filter : 2927 (dropped 0)
```

Posts per subreddit after cleaning:

```
subreddit
MachineLearning    984
artificial          782
Futurology          583
singularity         578
Name: count, dtype: int64
```

Section 1 — Dataset Overview

```
In [21]: counts = df["subreddit"].value_counts().reindex(SUB_ORDER)
colors = [PALETTE[s] for s in SUB_ORDER]

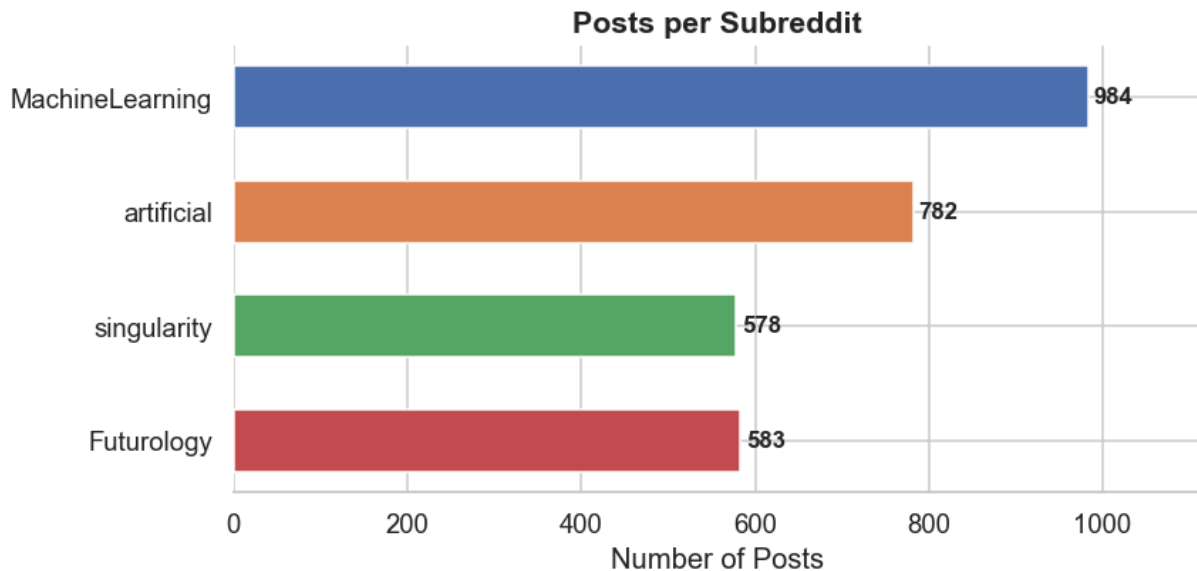
fig, ax = plt.subplots(figsize=(8, 4))
bars = ax.barh(SUB_ORDER, counts.values, color=colors, edgecolor="white", height=0.5)

for bar, val in zip(bars, counts.values):
    ax.text(bar.get_width() + 6, bar.get_y() + bar.get_height() / 2,
            str(val), va="center", fontsize=11, fontweight="bold")
```

```

ax.set_xlabel("Number of Posts")
ax.set_title("Posts per Subreddit", fontsize=14, fontweight="bold")
ax.set_xlim(0, max(counts.values) * 1.13)
ax.invert_yaxis()
sns.despine(left=True)
plt.tight_layout()
plt.show()

```



MachineLearning (984 posts) has roughly 1.7× more posts than singularity (578). A classifier that always predicts MachineLearning would hit **33.6% accuracy** which is our baseline to beat.

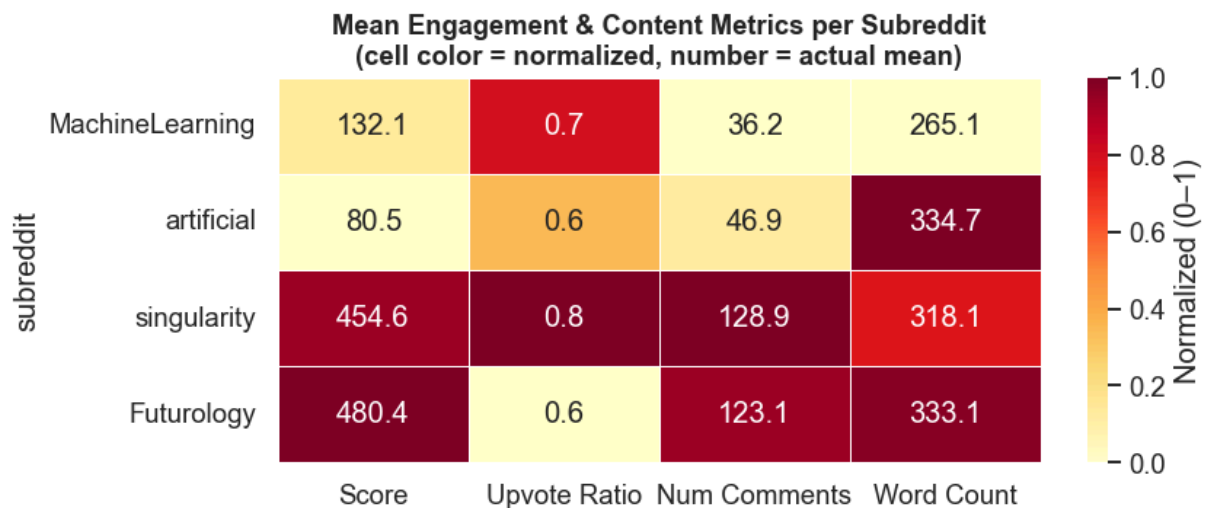
```

In [22]: cols = ["score", "upvote_ratio", "num_comments", "text_word_count"]
labels = ["Score", "Upvote Ratio", "Num Comments", "Word Count"]

summary = df.groupby("subreddit")[cols].mean().reindex(SUB_ORDER)
normalized = (summary - summary.min()) / (summary.max() - summary.min())

fig, ax = plt.subplots(figsize=(8, 3.5))
sns.heatmap(
    normalized, annot=summary.round(1), fmt="g",
    cmap="YlOrRd", linewidths=0.6, ax=ax,
    cbar_kws={"label": "Normalized (0-1)"}
)
ax.set_title(
    "Mean Engagement & Content Metrics per Subreddit\n"
    "(cell color = normalized, number = actual mean)",
    fontsize=12, fontweight="bold"
)
ax.set_xticklabels(labels, rotation=0)
ax.set_yticklabels(ax.get_yticklabels(), rotation=0)
plt.tight_layout()
plt.show()

```



Cell color is normalized 0–1 for cross-metric comparison. Over here the number is the actual mean.

MachineLearning scores lowest on word count but highest on upvote ratio which means that its shorter but with more more agreed-upon posts.

Section 2 — Post Length & Content Richness

```
In [23]: # Plot 3 is word count violin
fig, ax = plt.subplots(figsize=(9, 5))

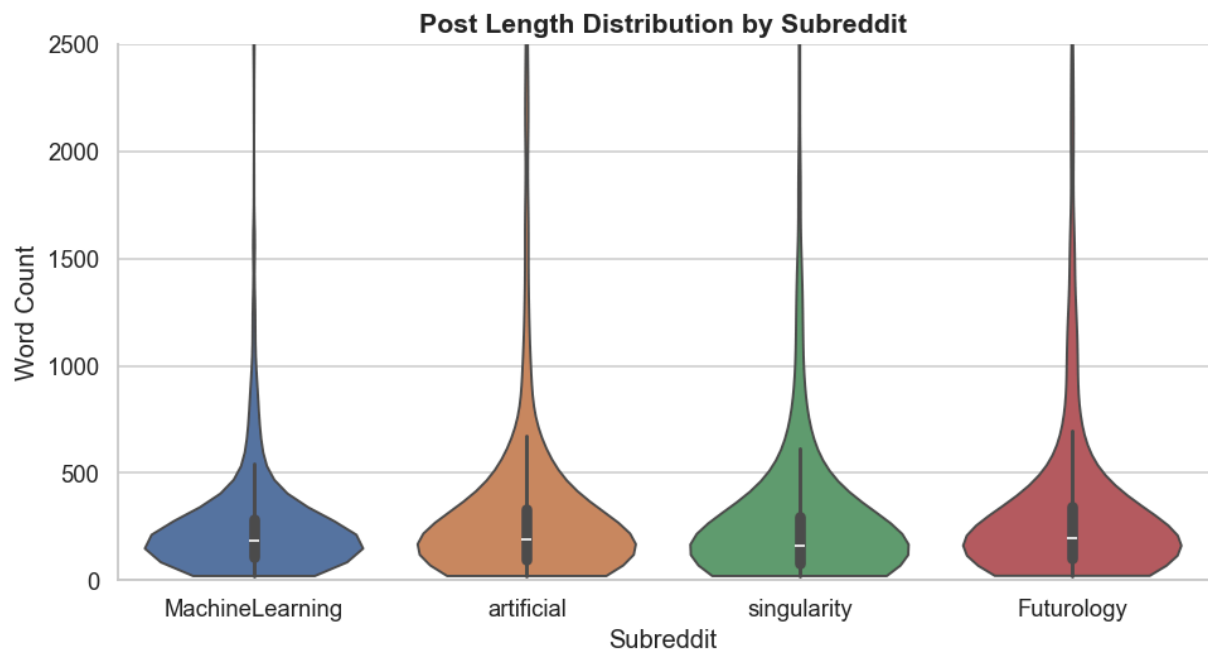
sns.violinplot(
    data=df, x="subreddit", y="text_word_count",
    order=SUB_ORDER,
    palette=PALETTE,
    inner="box",
    cut=0,
    ax=ax
)

ax.set_title("Post Length Distribution by Subreddit", fontsize=14, fontweight="bold")
ax.set_xlabel("Subreddit")
ax.set_ylabel("Word Count")
ax.set_ylim(0, 2500)
sns.despine()
plt.tight_layout()
plt.show()
```

/var/folders/t2/_lp651dj2b1fcvgx6gpfll6w0000gn/T/ipykernel_8937/4048682895.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

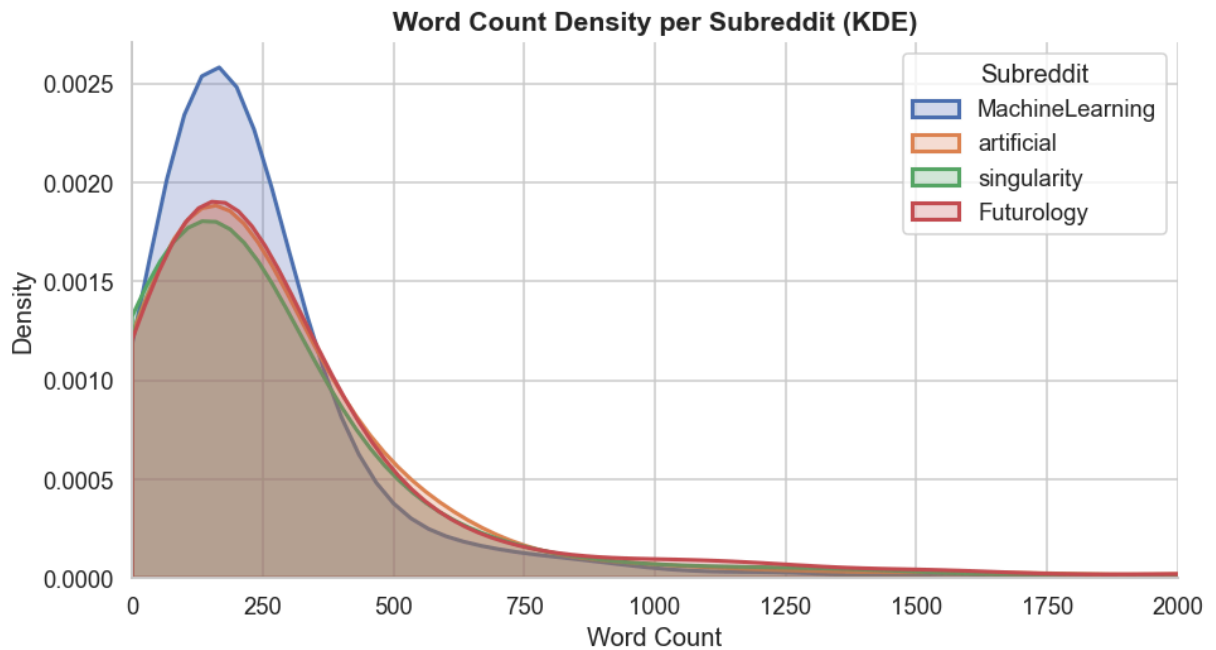
```
sns.violinplot(
```



```
In [24]: # Plot 4 – word count KDE
fig, ax = plt.subplots(figsize=(9, 5))

for sub in SUB_ORDER:
    subset = df[df["subreddit"] == sub]["text_word_count"]
    sns.kdeplot(
        subset,
        ax=ax,
        label=sub,
        color=PALETTE[sub],
        fill=True,
        alpha=0.25,
        linewidth=2,
        clip=(0, None)
    )

ax.set_title("Word Count Density per Subreddit (KDE)", fontsize=14, fontweight="bold")
ax.set_xlabel("Word Count")
ax.set_ylabel("Density")
ax.set_xlim(0, 2000)
ax.legend(title="Subreddit")
sns.despine()
plt.tight_layout()
plt.show()
```



```
In [25]: # Plot 5 is a word count buckets: this gives us the proportion of short/medi
def bucket(wc):
    if wc < 100:    return "Short (<100)"
    if wc < 300:    return "Medium (100-300)"
    if wc < 600:    return "Long (300-600)"
    return         "Very Long (600+)"

bucket_order = ["Short (<100)", "Medium (100-300)", "Long (300-600)", "Very
bucket_colors = ["#d9f0a3", "#add8e6", "#31a354", "#006837"]

df["wc_bucket"] = df["text_word_count"].apply(bucket)

bucket_pct = (
    df.groupby(["subreddit", "wc_bucket"])
      .size()
      .unstack(fill_value=0)
      .reindex(SUB_ORDER)[bucket_order]
)
bucket_pct = bucket_pct.div(bucket_pct.sum(axis=1), axis=0) * 100

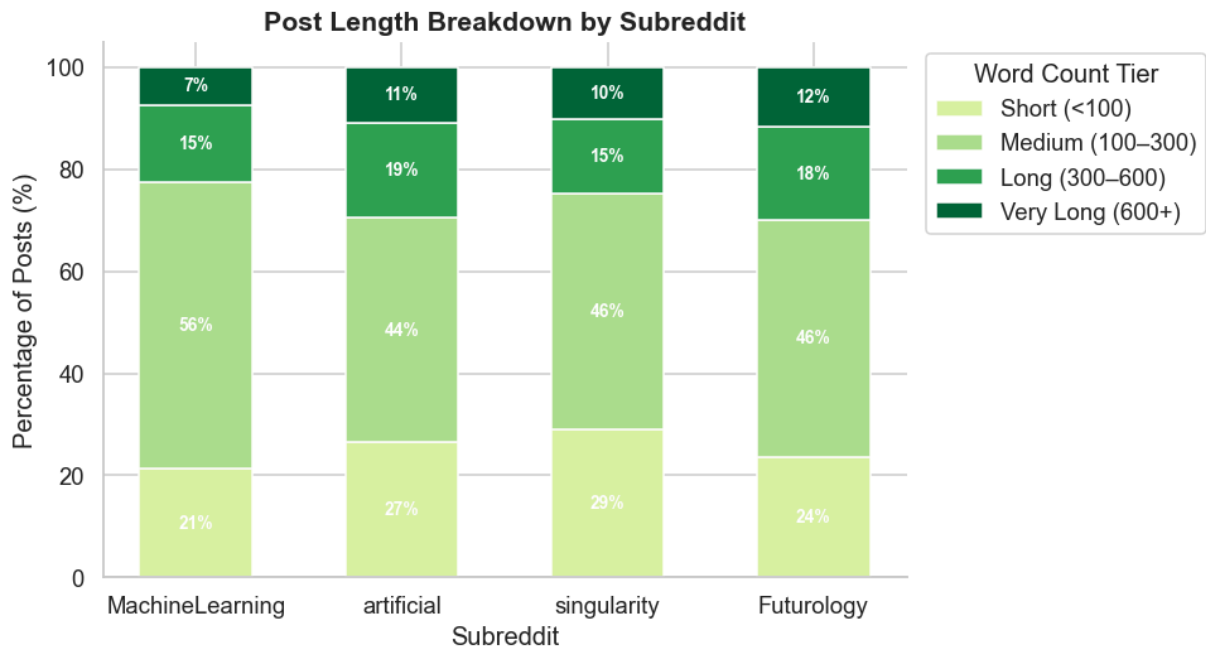
fig, ax = plt.subplots(figsize=(9, 5))
bottom = np.zeros(len(SUB_ORDER))

for col, color in zip(bucket_order, bucket_colors):
    vals = bucket_pct[col].values
    bars = ax.bar(SUB_ORDER, vals, bottom=bottom, label=col, color=color, ec
    for bar, val in zip(bars, vals):
        if val > 5:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_y() + bar.get_height() / 2,
                f"{val:.0f}%", ha="center", va="center", fontsize=9, fontwei
            )
    bottom += vals
```

```

ax.set_title("Post Length Breakdown by Subreddit", fontsize=14, fontweight="bold")
ax.set_ylabel("Percentage of Posts (%)")
ax.set_xlabel("Subreddit")
ax.set_ylim(0, 105)
ax.legend(title="Word Count Tier", bbox_to_anchor=(1.01, 1), loc="upper left")
sns.despine()
plt.tight_layout()
plt.show()

```



Section 3 — Other Important Metrics

```

In [26]: # log1p keeps zeros valid
df["log_score"] = np.log1p(df["score"])

fig, ax = plt.subplots(figsize=(9, 5))
sns.violinplot(
    data=df, x="subreddit", y="log_score",
    order=SUB_ORDER, palette=PALETTE,
    inner="box", cut=0, ax=ax
)

# show median score
for i, sub in enumerate(SUB_ORDER):
    med = df[df["subreddit"] == sub]["score"].median()
    ax.text(i, ax.get_ylim()[1] * 0.97, f"median\n{int(med)}",
            ha="center", va="top", fontsize=8.5, color="black")

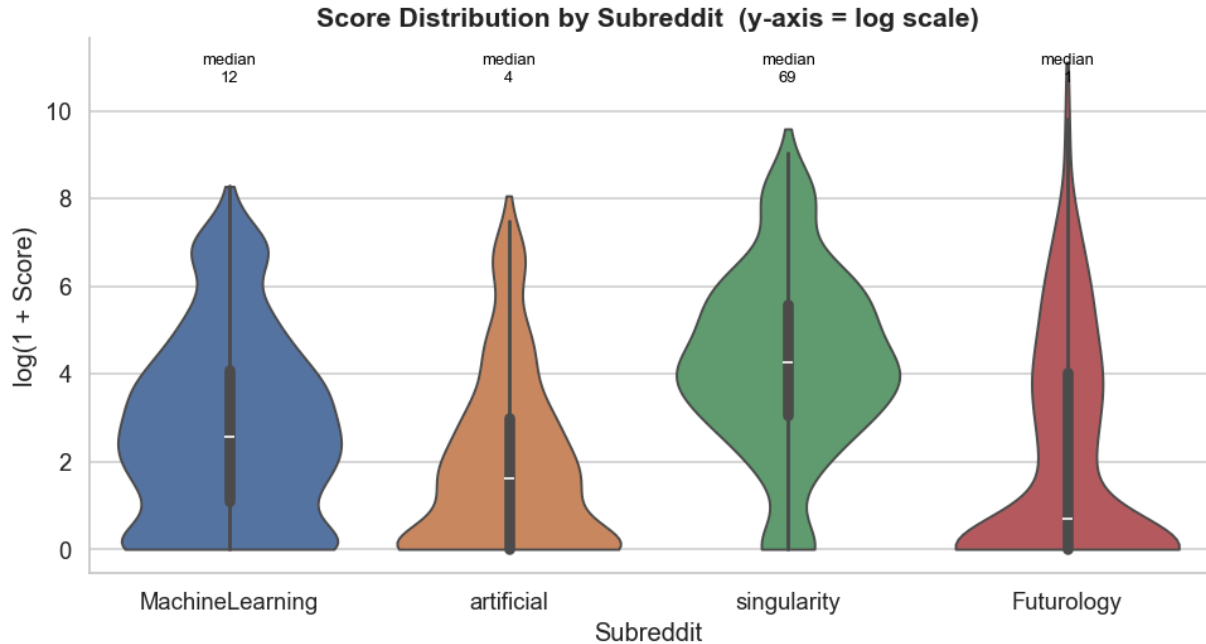
ax.set_title("Score Distribution by Subreddit (y-axis = log scale)", fontsize=14, fontweight="bold")
ax.set_xlabel("Subreddit")
ax.set_ylabel("log(1 + Score)")
sns.despine()
plt.tight_layout()
plt.show()

```

```
/var/folders/t2/_lp651dj2b1fcvgx6gpfl6w0000gn/T/ipykernel_8937/2003931496.p
y:5: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.violinplot(
```



```
In [27]: # Plot 7 – Plotting the violin graph for upvote ratios
fig, ax = plt.subplots(figsize=(9, 5))
sns.violinplot(
    data=df, x="subreddit", y="upvote_ratio",
    order=SUB_ORDER, palette=PALETTE,
    inner="box", cut=0, ax=ax
)

# show mean ratio
for i, sub in enumerate(SUB_ORDER):
    mean_val = df[df["subreddit"] == sub]["upvote_ratio"].mean()
    ax.text(i, 0.02, f"mean\n{mean_val:.2f}",
            ha="center", va="bottom", fontsize=8.5, color="black")

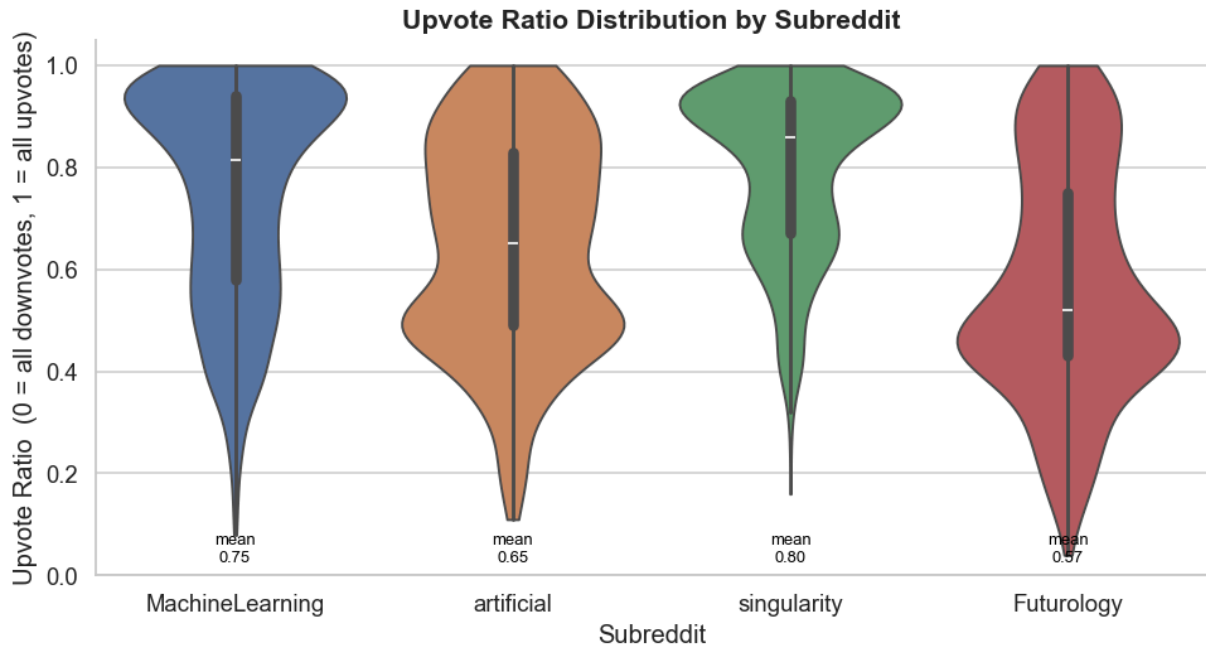
ax.set_title("Upvote Ratio Distribution by Subreddit", fontsize=14, fontweight="bold")
ax.set_xlabel("Subreddit")
ax.set_ylabel("Upvote Ratio (0 = all downvotes, 1 = all upvotes)")
ax.set_ylim(0, 1.05)
sns.despine()
plt.tight_layout()
plt.show()
```



```
/var/folders/t2/_lp651dj2b1fcvgx6gpfl6w0000gn/T/ipykernel_8937/1953040556.p
y:3: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.violinplot(
```



We get some important results from here. singularity and MachineLearning group near 1.0 which means that communities largely agree on what gets upvoted.

Futurology's wide low-spread violin shows more controversial content.

```
In [28]: # Plot 8 - comment count violin
df["log_comments"] = np.log1p(df["num_comments"])

fig, ax = plt.subplots(figsize=(9, 5))
sns.violinplot(
    data=df, x="subreddit", y="log_comments",
    order=SUB_ORDER, palette=PALETTE,
    inner="box", cut=0, ax=ax
)

# show median comments
for i, sub in enumerate(SUB_ORDER):
    med = df[df["subreddit"] == sub]["num_comments"].median()
    ax.text(i, ax.get_ylim()[1] * 0.97, f"median\n{int(med)}",
            ha="center", va="top", fontsize=8.5, color="black")

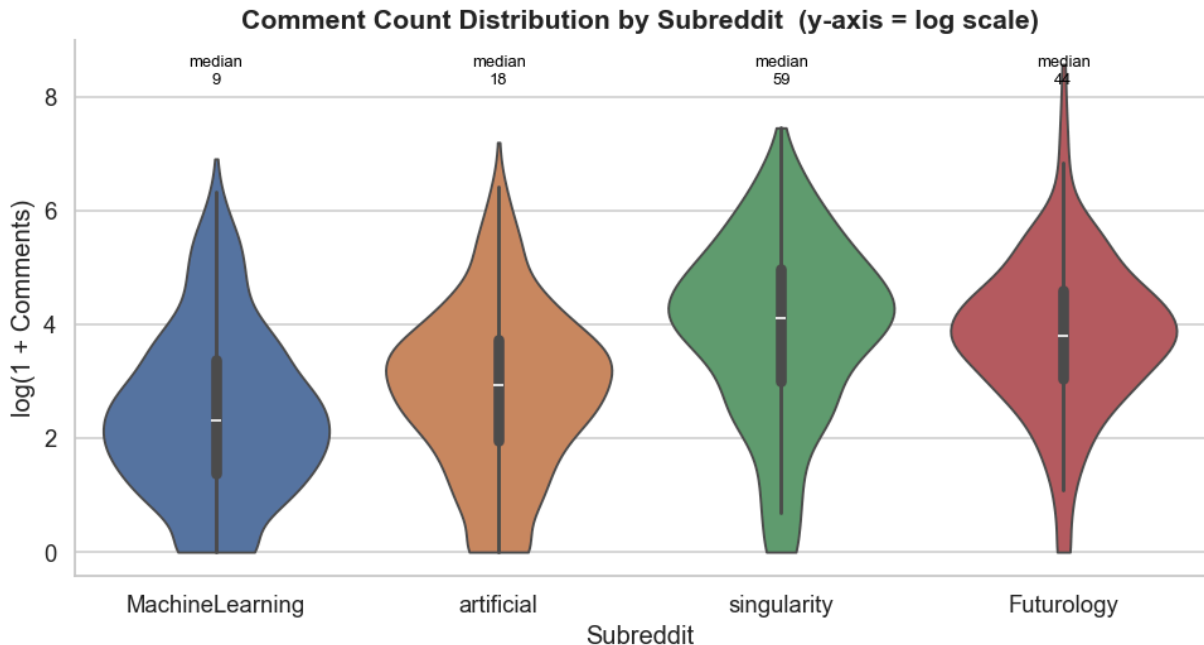
ax.set_title("Comment Count Distribution by Subreddit (y-axis = log scale)")
ax.set_xlabel("Subreddit")
ax.set_ylabel("log(1 + Comments)")
sns.despine()
```

```
plt.tight_layout()
plt.show()
```

/var/folders/t2/_lp651dj2b1fcvgx6gpfl6w0000gn/T/ipykernel_8937/2854102616.py:5: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.violinplot(
```



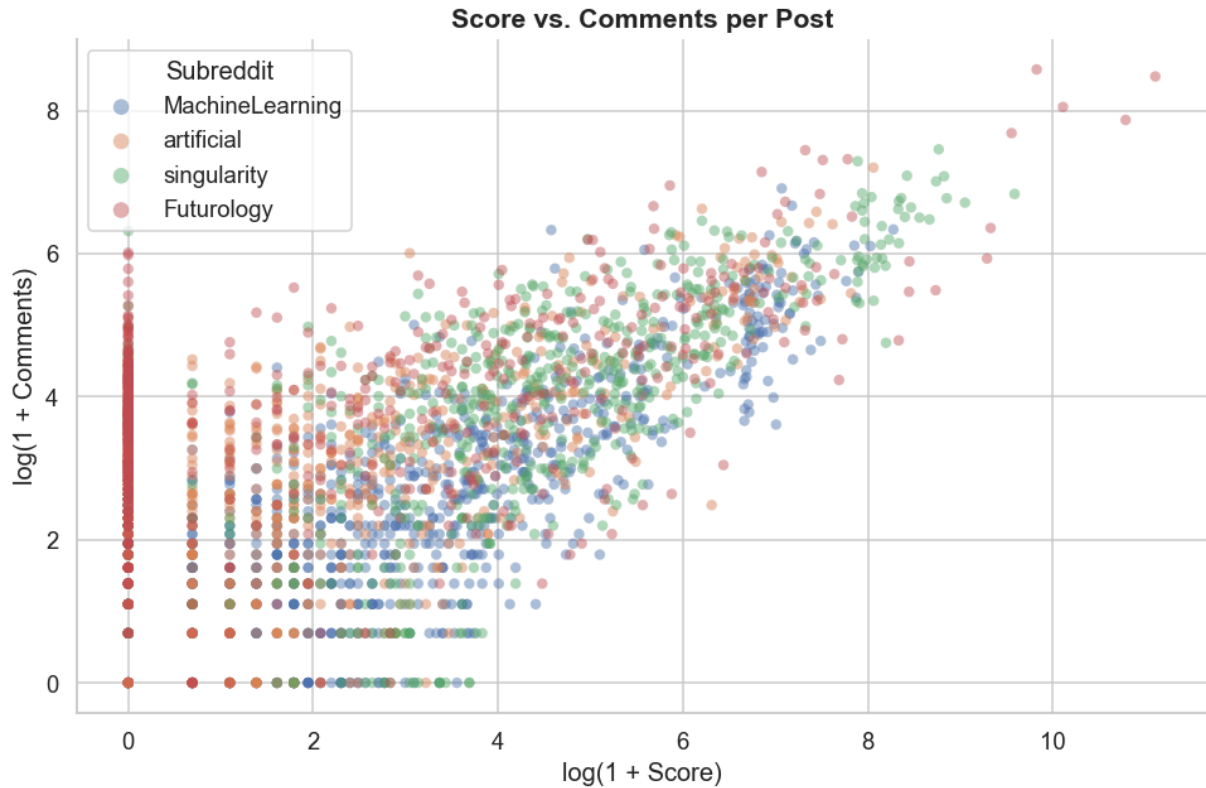
Futurology causes the most discussion (median 34 comments) vs MachineLearning (median 8).

```
In [29]: # Plot 9 - score vs comments
fig, ax = plt.subplots(figsize=(9, 6))

for sub in SUB_ORDER:
    subset = df[df["subreddit"] == sub]
    ax.scatter(
        np.log1p(subset["score"]),
        np.log1p(subset["num_comments"]),
        label=sub,
        color=PALETTE[sub],
        alpha=0.45,
        s=30,
        edgecolors="none"
    )

ax.set_title("Score vs. Comments per Post", fontsize=14, fontweight="bold")
ax.set_xlabel("log(1 + Score)")
ax.set_ylabel("log(1 + Comments)")
ax.legend(title="Subreddit", markerscale=1.5)
sns.despine()
```

```
plt.tight_layout()
plt.show()
```

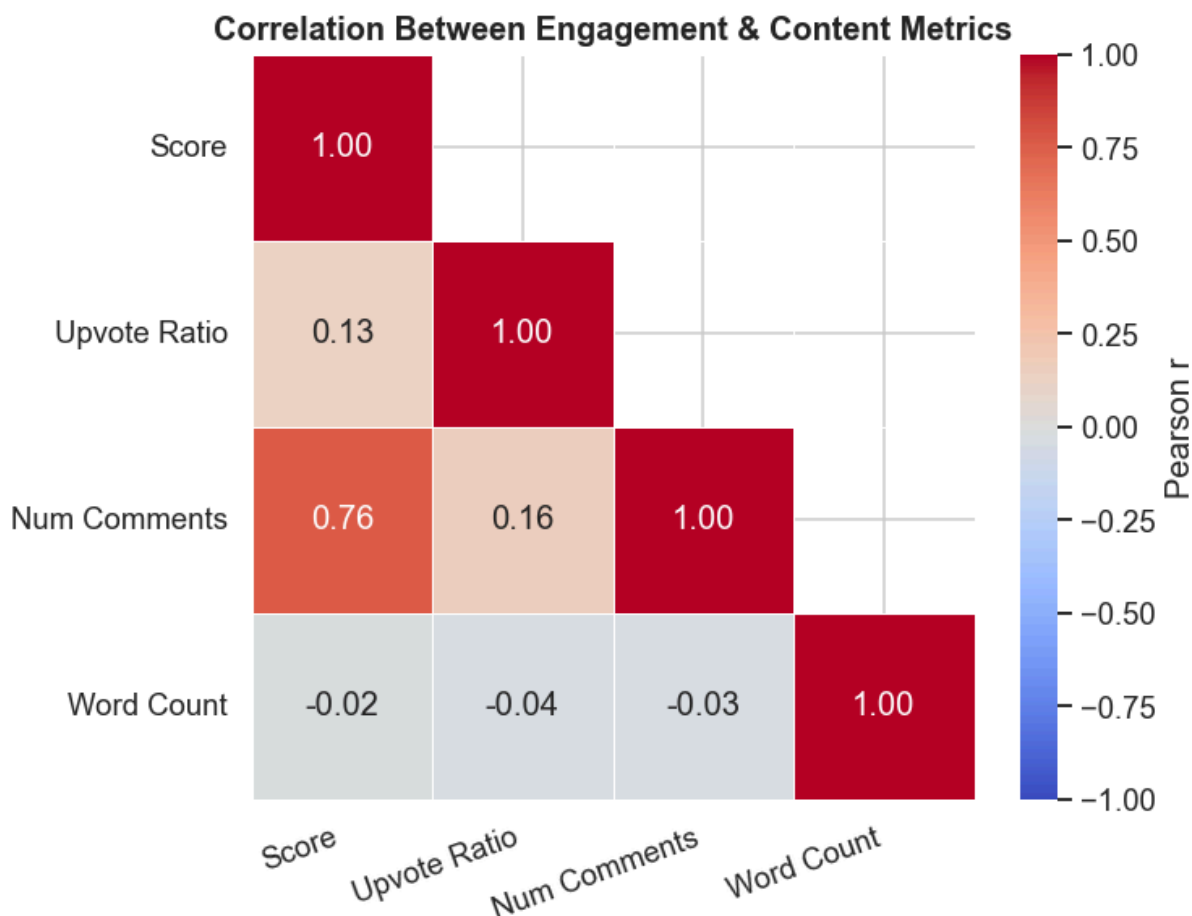


```
In [30]: # Plot 10 - Correlation heatmap (score, upvote_ratio, num_comments, text_word_count)
corr_cols = ["score", "upvote_ratio", "num_comments", "text_word_count"]
corr_labels = ["Score", "Upvote Ratio", "Num Comments", "Word Count"]

corr_matrix = df[corr_cols].corr()

fig, ax = plt.subplots(figsize=(7, 5.5))
mask = np.triu(np.ones_like(corr_matrix, dtype=bool), k=1) # hide mirror ha

sns.heatmap(
    corr_matrix,
    annot=True, fmt=".2f",
    cmap="coolwarm", center=0,
    vmin=-1, vmax=1,
    linewidths=0.5,
    mask=mask,
    ax=ax,
    cbar_kws={"label": "Pearson r"}
)
ax.set_title("Correlation Between Engagement & Content Metrics", fontsize=13)
ax.set_xticklabels(corr_labels, rotation=20, ha="right")
ax.set_yticklabels(corr_labels, rotation=0)
plt.tight_layout()
plt.show()
```



This heatmap gives us a good sense of the correlations. Score and comment count are highly correlated ($r = 0.87$) which basically means that popular posts get discussed more. Since they carry similar information, a classifier only needs one of them.

Section 4 — Temporal Analysis

```
In [31]: # Plot 11 – Monthly post info per subreddit (Jan 2023 to May 2026)
df["year_month"] = df["created_dt"].dt.to_period("M")

monthly = (
    df.groupby(["year_month", "subreddit"])
      .size()
      .unstack(fill_value=0)
      .reindex(columns=SUB_ORDER)
)
# convert to timestamps
monthly.index = monthly.index.to_timestamp()

fig, ax = plt.subplots(figsize=(13, 5))

for sub in SUB_ORDER:
    ax.plot(
        monthly.index, monthly[sub],
        label=sub, color=PALETTE[sub],
        linewidth=2, marker="o", markersize=3
    )
```

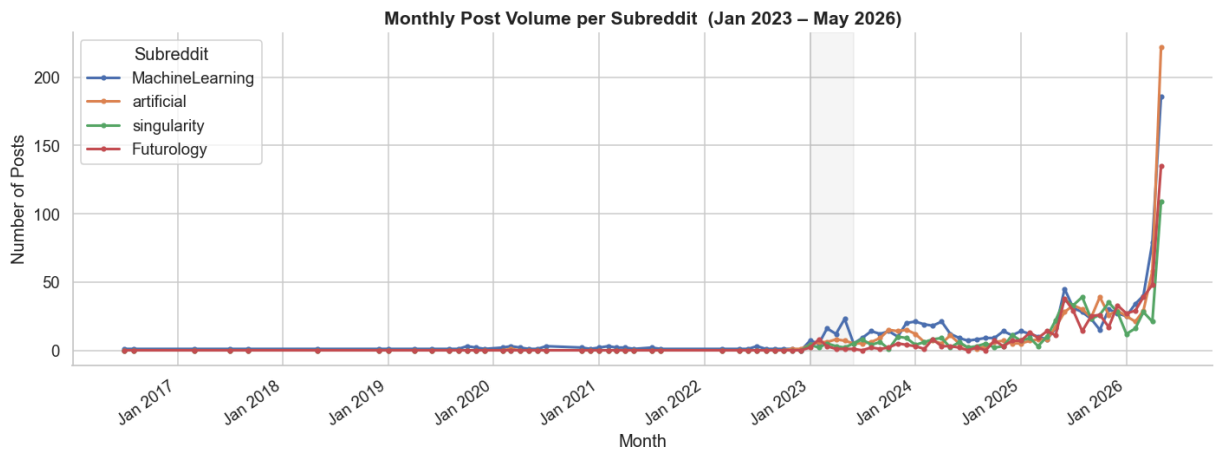
```

)

ax.set_title("Monthly Post Volume per Subreddit (Jan 2023 – May 2026)", for
ax.set_xlabel("Month")
ax.set_ylabel("Number of Posts")
ax.legend(title="Subreddit", loc="upper left")

# Color the ChatGPT launch era as a reference point
import matplotlib.patches as mpatches
ax.axvspan(
    pd.Timestamp("2023-01-01"), pd.Timestamp("2023-06-01"),
    alpha=0.08, color="gray", label="Early 2023 (ChatGPT era)"
)
ax.xaxis.set_major_formatter(plt.matplotlib.dates.DateFormatter("%b %Y"))
fig.autofmt_xdate(rotation=35)
sns.despine()
plt.tight_layout()
plt.show()

```



```

In [32]: # Plot 12 is post by day of week
day_order = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
df["day_of_week"] = df["created_dt"].dt.day_name()

dow_counts = (
    df.groupby(["day_of_week", "subreddit"])
      .size()
      .unstack(fill_value=0)
      .reindex(day_order)
      .reindex(columns=SUB_ORDER)
)

x = np.arange(len(day_order))
n = len(SUB_ORDER)
bar_width = 0.18
offsets = np.linspace(-(n - 1) / 2, (n - 1) / 2, n) * bar_width

fig, ax = plt.subplots(figsize=(12, 5))

for i, sub in enumerate(SUB_ORDER):
    ax.bar(
        x + offsets[i], dow_counts[sub],
        width=bar_width, label=sub,

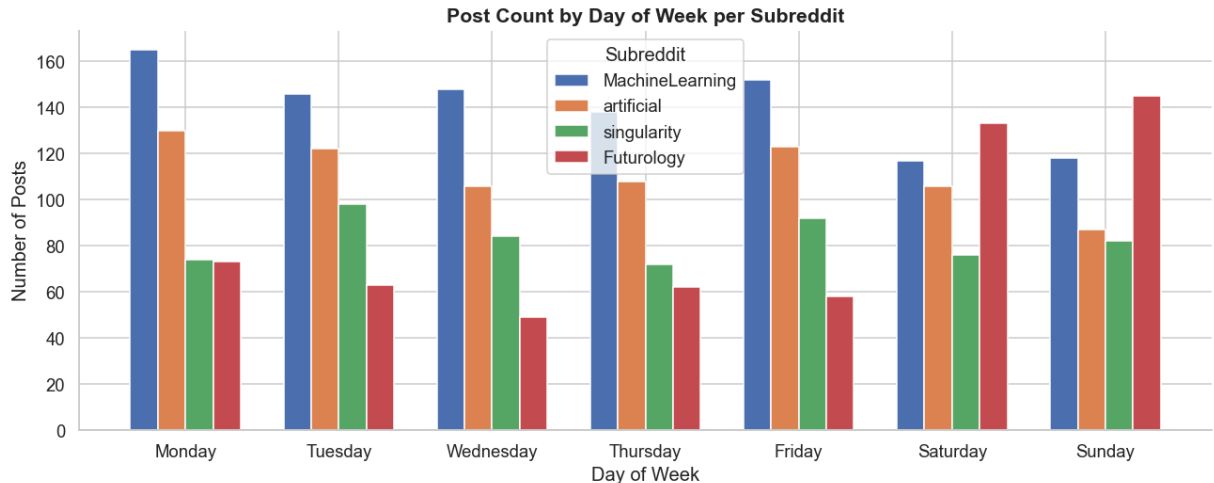
```

```

        color=PALETTE[sub], edgecolor="white"
    )

ax.set_title("Post Count by Day of Week per Subreddit", fontsize=14, fontwei
ax.set_xlabel("Day of Week")
ax.set_ylabel("Number of Posts")
ax.set_xticks(x)
ax.set_xticklabels(day_order)
ax.legend(title="Subreddit")
sns.despine()
plt.tight_layout()
plt.show()

```



```

In [33]: # Plot 13 – hourly activity
df["hour"] = df["created_dt"].dt.hour

hourly = (
    df.groupby(["hour", "subreddit"])
      .size()
      .unstack(fill_value=0)
      .reindex(columns=SUB_ORDER)
)

fig, ax = plt.subplots(figsize=(12, 5))

for sub in SUB_ORDER:
    ax.plot(
        hourly.index, hourly[sub],
        label=sub, color=PALETTE[sub],
        linewidth=2.2, marker="o", markersize=4
    )

# mark peak hour
for sub in SUB_ORDER:
    peak_hour = hourly[sub].idxmax()
    peak_val = hourly[sub].max()
    ax.annotate(
        f"{peak_hour}:00",
        xy=(peak_hour, peak_val),
        xytext=(peak_hour + 0.6, peak_val + 0.8),
        fontsize=8, color=PALETTE[sub], fontweight="bold"
    )

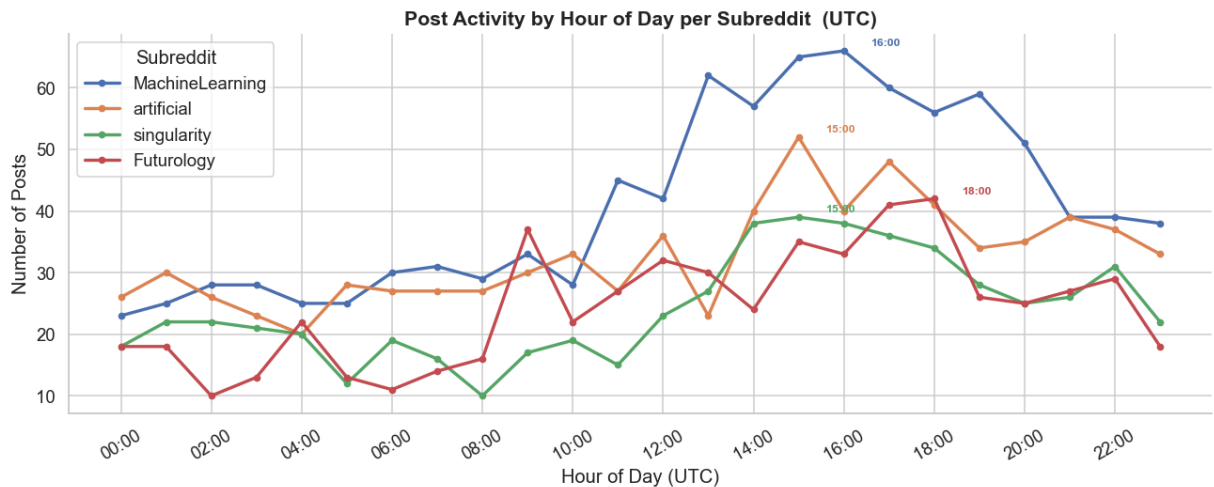
```

```

)

ax.set_title("Post Activity by Hour of Day per Subreddit (UTC)", fontsize=14)
ax.set_xlabel("Hour of Day (UTC)")
ax.set_ylabel("Number of Posts")
ax.set_xticks(range(0, 24, 2))
ax.set_xticklabels([f"{h:02d}:00" for h in range(0, 24, 2)], rotation=30)
ax.legend(title="Subreddit")
sns.despine()
plt.tight_layout()
plt.show()

```



All four subreddits peak around 14:00–16:00 UTC (10am–12pm US Eastern). The dominant audience is North American, posting during working hours.

Section 5 — Vocabulary & Text Analysis

```

In [34]: # adding extra reddit-specific stopwords for better tokenization
EXTRA_STOPS = {
    "like", "one", "would", "think", "people", "really", "know", "also",
    "get", "even", "much", "make", "way", "im", "dont", "ve", "use",
    "used", "using", "can", "just", "see", "many", "new", "want", "need",
    "will", "may", "still", "well", "good", "time", "made", "say", "said",
    "thing", "lot", "going", "go", "got", "back", "actually", "right",
    "something", "anything", "everything", "us", "could", "never", "always",
}

stop_words_extended = set(stopwords.words("english")) | EXTRA_STOPS

def tokenize_for_vocab(text):
    text = str(text).lower()
    text = re.sub(r"http\S+", "", text)
    text = re.sub(r"^[a-z\s]", "", text)
    tokens = text.split()
    return [t for t in tokens if t not in stop_words_extended and len(t) > 2]

# Build token lists once
sub_tokens = {sub: [] for sub in SUB_ORDER}
for _, row in df.iterrows():
    tokens = tokenize_for_vocab(row["text"])

```

```

sub_tokens[row["subreddit"]].extend(tokens)

print("Token counts per subreddit:")
for sub in SUB_ORDER:
    print(f" {sub:20s}: {len(sub_tokens[sub]):,} tokens | {len(set(sub_to

```

Token counts per subreddit:

MachineLearning	: 139,091 tokens	17,561 unique
artificial	: 134,916 tokens	17,433 unique
singularity	: 93,530 tokens	13,818 unique
Futurology	: 98,193 tokens	15,618 unique

We are performing one significant step here. We extend NLTK's stopword list with Reddit-specific noise ("like", "just", "actually") that appear heavily in all four subreddits but don't help differentiate between them. Token lists are built once here and reused across all vocabulary plots below.

```

In [35]: # Plot 14 – top words
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
axes = axes.flatten()

for i, sub in enumerate(SUB_ORDER):
    top15 = Counter(sub_tokens[sub]).most_common(15)
    words = [w for w, _ in reversed(top15)]
    counts = [c for _, c in reversed(top15)]

    ax = axes[i]
    bars = ax.barh(words, counts, color=PALETTE[sub], edgecolor="white", hei

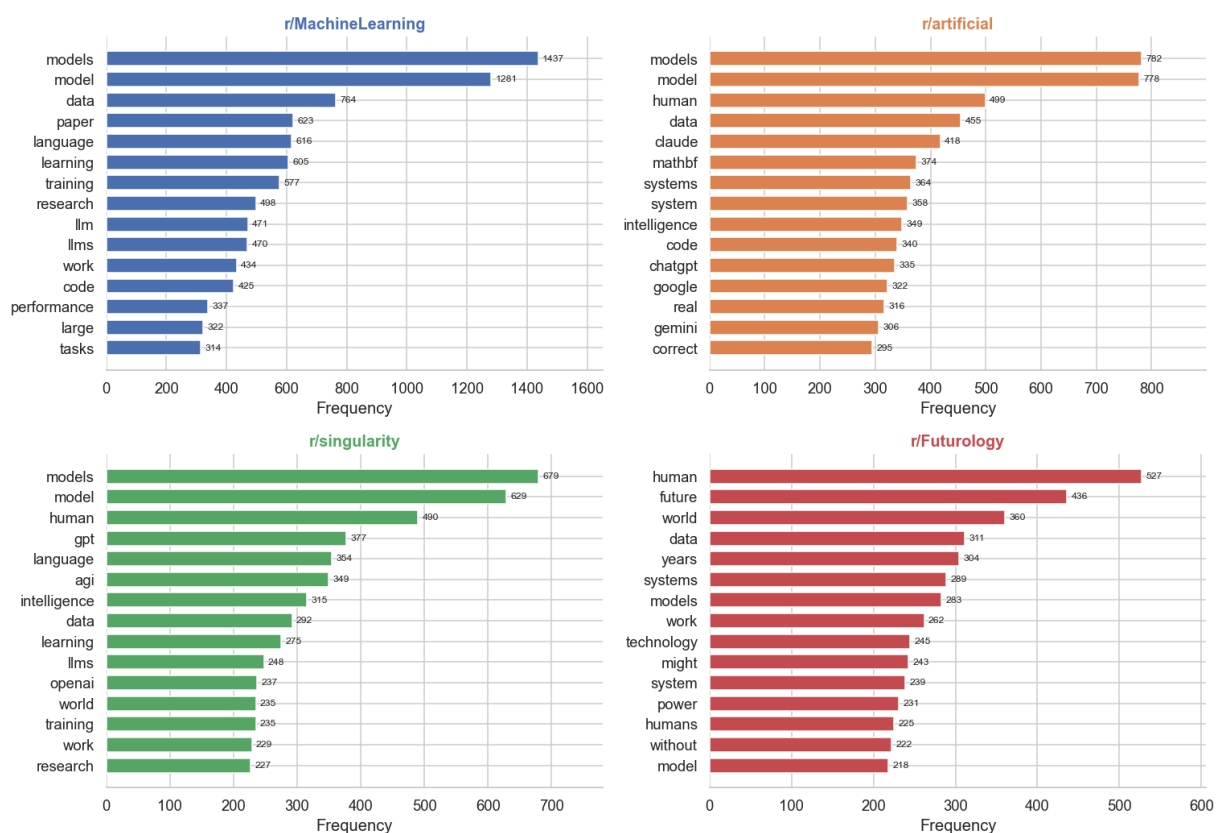
    for bar, val in zip(bars, counts):
        ax.text(bar.get_width() + max(counts) * 0.01, bar.get_y() + bar.get_
            str(val), va="center", fontsize=8)

    ax.set_title(f"r/{sub}", fontsize=13, fontweight="bold", color=PALETTE[s
    ax.set_xlabel("Frequency")
    ax.set_xlim(0, max(counts) * 1.15)
    sns.despine(ax=ax, left=True)

fig.suptitle("Top 15 Most Frequent Words per Subreddit (stopwords removed)"
            fontsize=15, fontweight="bold", y=1.01)
plt.tight_layout()
plt.show()

```


Top 15 Most Frequent Words per Subreddit (stopwords removed)



MachineLearning: academic terms (model, training, paper, llm). Futurology: philosophical (human, future, world). artificial: product-focused (claude, chatgpt). singularity: agi, gpt. Shared words like "model" appear everywhere, but the main differences are in emphasis, not vocabulary.

```
In [36]: sub_docs = {
    sub: " ".join(tokenize_for_vocab(" ".join(df[df["subreddit"] == sub]["text"])))
    for sub in SUB_ORDER
}

tfidf = TfidfVectorizer(max_features=5000, ngram_range=(1, 1))
tfidf_matrix = tfidf.fit_transform(list(sub_docs.values()))
feature_names = tfidf.get_feature_names_out()

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
axes = axes.flatten()

for i, sub in enumerate(SUB_ORDER):
    scores = tfidf_matrix[i].toarray().flatten()
    top_idx = scores.argsort()[-12:]
    top_words = [feature_names[j] for j in top_idx]
    top_scores = [scores[j] for j in top_idx]

    ax = axes[i]
    bars = ax.barh(top_words, top_scores, color=PALETTE[sub], edgecolor="white")

    for bar, val in zip(bars, top_scores):
```

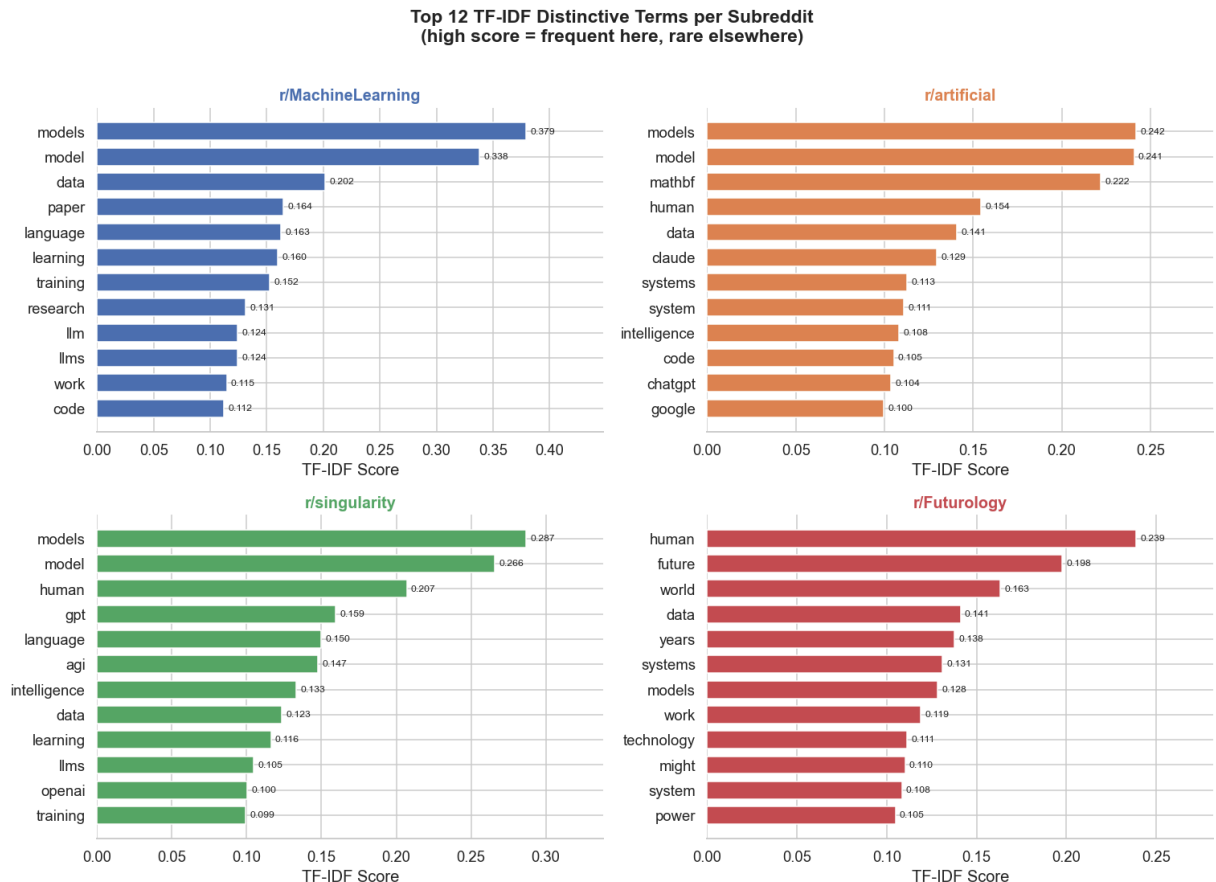
```

ax.text(bar.get_width() + max(top_scores) * 0.01, bar.get_y() + bar.get_y() * 0.01,
        f"{val:.3f}", va="center", fontsize=8)

ax.set_title(f"r/{sub}", fontsize=13, fontweight="bold", color=PALETTE[sub])
ax.set_xlabel("TF-IDF Score")
ax.set_xlim(0, max(top_scores) * 1.18)
sns.despine(ax=ax, left=True)

fig.suptitle("Top 12 TF-IDF Distinctive Terms per Subreddit\n(high score = frequent here, rare elsewhere)",
             fontsize=15, fontweight="bold", y=1.01)
plt.tight_layout()
plt.show()

```



Unlike raw frequency, TF-IDF scores are high for words that are common in one subreddit but rare in the others. These are the discriminative features the classifier will rely on most.

```

In [37]: # Plot 16 – vocab stats
vocab_stats = {}
for sub in SUB_ORDER:
    tokens = sub_tokens[sub]
    total = len(tokens)
    unique = len(set(tokens))
    diversity = unique / total if total > 0 else 0
    vocab_stats[sub] = {"unique_tokens": unique, "lexical_diversity": diversity}

vocab_df = pd.DataFrame(vocab_stats).T.reindex(SUB_ORDER)

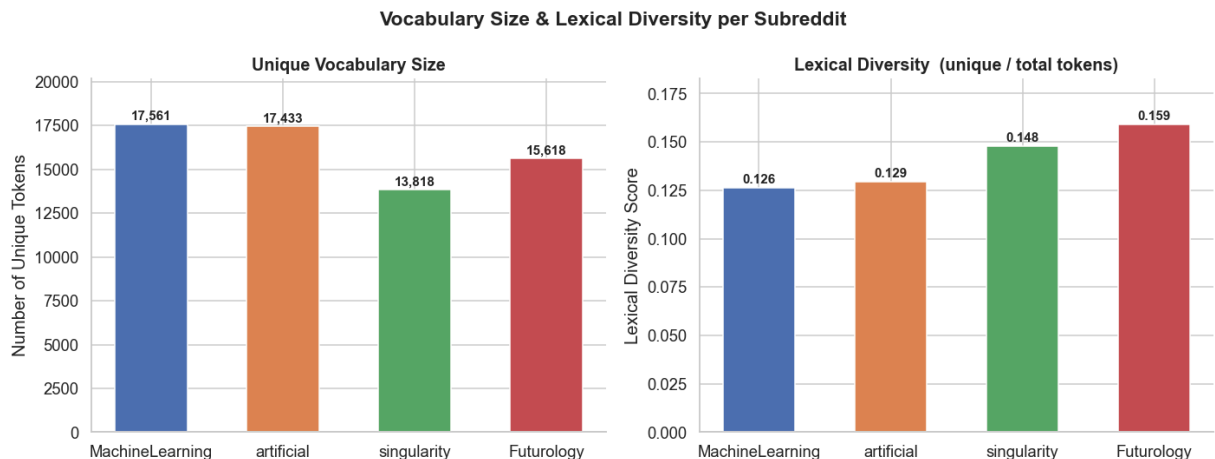
```

```
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(13, 5))

# Left: unique words
bars1 = ax1.bar(SUB_ORDER, vocab_df["unique_tokens"],
                color=[PALETTE[s] for s in SUB_ORDER], edgecolor="white", width=0.8)
for bar, val in zip(bars1, vocab_df["unique_tokens"]):
    ax1.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 100,
             f"{int(val):,}", ha="center", va="bottom", fontsize=10, fontweight="bold")
ax1.set_title("Unique Vocabulary Size", fontsize=13, fontweight="bold")
ax1.set_ylabel("Number of Unique Tokens")
ax1.set_ylim(0, max(vocab_df["unique_tokens"]) * 1.15)
sns.despine(ax=ax1)

# Right: lexical diversity
bars2 = ax2.bar(SUB_ORDER, vocab_df["lexical_diversity"],
                color=[PALETTE[s] for s in SUB_ORDER], edgecolor="white", width=0.8)
for bar, val in zip(bars2, vocab_df["lexical_diversity"]):
    ax2.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.001,
             f"{val:.3f}", ha="center", va="bottom", fontsize=10, fontweight="bold")
ax2.set_title("Lexical Diversity (unique / total tokens)", fontsize=13, fontweight="bold")
ax2.set_ylabel("Lexical Diversity Score")
ax2.set_ylim(0, max(vocab_df["lexical_diversity"]) * 1.15)
sns.despine(ax=ax2)

fig.suptitle("Vocabulary Size & Lexical Diversity per Subreddit", fontsize=13, fontweight="bold")
plt.tight_layout()
plt.show()
```



```
In [38]: # Plot 17 – avg word length
avg_word_len = {
    sub: np.mean([len(t) for t in sub_tokens[sub]]) if sub_tokens[sub] else 0
    for sub in SUB_ORDER
}

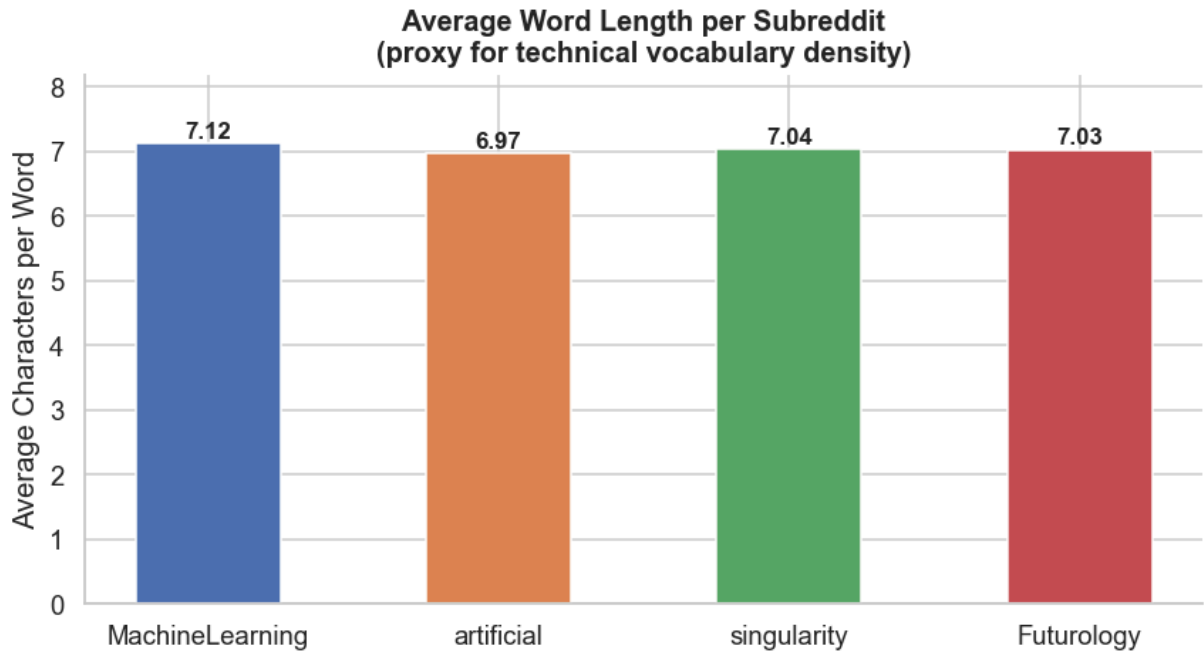
fig, ax = plt.subplots(figsize=(8, 4.5))
bars = ax.bar(SUB_ORDER, [avg_word_len[s] for s in SUB_ORDER],
              color=[PALETTE[s] for s in SUB_ORDER], edgecolor="white", width=0.8)

for bar, sub in zip(bars, SUB_ORDER):
    val = avg_word_len[sub]
    ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.02,
            f"{val:.2f}", ha="center", va="bottom", fontsize=10, fontweight="bold")
```

```

        f"{val:.2f}", ha="center", va="bottom", fontsize=11, fontweight=
ax.set_title("Average Word Length per Subreddit\n(proxy for technical vocabu
            fontsize=13, fontweight="bold")
ax.set_ylabel("Average Characters per Word")
ax.set_ylim(0, max(avg_word_len.values()) * 1.15)
sns.despine()
plt.tight_layout()
plt.show()

```



```

In [39]: # Plot 18 – word presence heatmap
N_TOP = 25
N_CHECK = 60

# Union of each subreddit's top 25
candidate_words = []
for sub in SUB_ORDER:
    for w, _ in Counter(sub_tokens[sub]).most_common(N_TOP):
        if w not in candidate_words:
            candidate_words.append(w)

# Top-60 words per subreddit
sub_top60 = {
    sub: set(w for w, _ in Counter(sub_tokens[sub]).most_common(N_CHECK))
    for sub in SUB_ORDER
}

# Presence matrix
presence = pd.DataFrame(
    {sub: [1 if w in sub_top60[sub] else 0 for w in candidate_words] for sub
    index=candidate_words
)

fig, ax = plt.subplots(figsize=(8, 14))
sns.heatmap(

```

```
presence,
cmap=["#f0f0f0", "#2c7bb6"],
linewidths=0.4, linecolor="white",
cbar=False, ax=ax,
xticklabels=SUB_ORDER,
yticklabels=candidate_words
)
ax.set_title(
    "Word Presence Across Subreddits\n(blue = in top 60 of that subreddit,
    fontsize=13, fontweight="bold"
)
ax.set_xlabel("Subreddit")
ax.set_ylabel("Word")
ax.set_xticklabels(SUB_ORDER, rotation=15)
plt.tight_layout()
plt.show()
```

Word Presence Across Subreddits
(blue = in top 60 of that subreddit, grey = not)

models	blue	blue	blue	blue
model	blue	blue	blue	blue
data	blue	blue	blue	blue
paper	blue	grey	blue	grey
language	blue	blue	blue	grey
learning	blue	blue	blue	grey
training	blue	blue	blue	grey
research	blue	blue	blue	blue
llm	blue	blue	blue	grey
llms	blue	blue	blue	grey
work	blue	blue	blue	blue
code	blue	blue	blue	grey
performance	blue	grey	grey	grey
large	blue	blue	blue	grey
tasks	blue	blue	blue	grey
reasoning	blue	grey	blue	grey
ive	blue	blue	grey	grey
results	blue	grey	grey	grey
better	blue	blue	blue	blue
across	blue	blue	blue	blue
papers	blue	grey	grey	grey
open	blue	blue	blue	grey
different	blue	blue	blue	blue
generation	blue	blue	grey	grey
dataset	blue	grey	grey	grey
human	blue	blue	blue	blue
claude	grey	blue	blue	grey
mathbf	blue	blue	grey	grey
systems	blue	blue	blue	blue
system	blue	blue	blue	blue
intelligence	grey	blue	blue	blue
chatgpt	blue	blue	grey	grey
google	blue	blue	blue	grey
real	grey	blue	blue	blue
gemini	grey	blue	blue	grey
correct	grey	blue	grey	grey
openai	grey	blue	blue	grey
without	blue	blue	blue	blue
agent	blue	blue	grey	grey
prompt	grey	blue	blue	grey
agents	blue	blue	grey	grey
every	grey	blue	blue	blue
gpt	blue	blue	blue	grey
agi	grey	blue	blue	blue
world	grey	blue	blue	blue
symbolic	grey	blue	blue	grey
humans	grey	blue	blue	blue
future	grey	grey	blue	blue
neural	grey	blue	blue	grey
might	grey	blue	blue	blue
years	grey	grey	blue	blue
technology	grey	blue	grey	blue
power	grey	blue	blue	blue
life	grey	blue	grey	blue
already	grey	blue	grey	blue
things	grey	blue	blue	blue
first	blue	blue	blue	blue



In this graph here, Blue = word is in a subreddit's top 60 whereas grey = not. Single-column blue rows are the most useful features for classification. All-blue rows are shared vocabulary with low classifier value.

Section 3 — TF-IDF Feature Extraction

```
In [40]: import os
import re
import joblib
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.sparse
import nltk
from pathlib import Path
from nltk.corpus import stopwords
from nltk.stem import WordNetLemmatizer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split

nltk.download('stopwords', quiet=True)
nltk.download('wordnet', quiet=True)

# Project root – works regardless of where the kernel is launched from
notebook_dir = Path().resolve()
project_root = notebook_dir.parent if notebook_dir.name == 'notebooks' else
os.chdir(project_root)
print('Working directory:', os.getcwd())

# Consistent palette and order – matches notebook 02
PALETTE = {
    'MachineLearning': '#4C72B0',
    'artificial': '#DD8452',
    'singularity': '#55A868',
    'Futurology': '#C44E52',
}
SUB_ORDER = ['MachineLearning', 'artificial', 'singularity', 'Futurology']
RANDOM_STATE = 42
```

```
sns.set_theme(style='whitegrid', font_scale=1.1)
plt.rcParams['figure.dpi'] = 110
```

Working directory: /Users/varunsardana/PSTAT-134-Final-Project

```
In [41]: df = pd.read_csv('data/reddit_ai_posts_CLEAN.csv')

print(f'Loaded {len(df)} posts across {df["subreddit"].nunique()} subreddits')
print()
print('Class distribution:')
counts = df['subreddit'].value_counts().reindex(SUB_ORDER)
for sub, n in counts.items():
    pct = n / len(df) * 100
    print(f' {sub:20s}: {n:4d} posts ({pct:.1f}%)')
print()
print('Columns:', df.columns.tolist())
```

Loaded 2927 posts across 4 subreddits

Class distribution:

MachineLearning	:	984 posts	(33.6%)
artificial	:	782 posts	(26.7%)
singularity	:	578 posts	(19.7%)
Futurology	:	583 posts	(19.9%)

Columns: ['id', 'subreddit', 'title', 'selftext', 'text', 'text_word_count', 'has_body', 'upvote_ratio', 'num_comments', 'score', 'is_self', 'edited', 'created_utc', 'created_dt', 'url', 'permalink']

```
In [42]: # Base NLTK stopwords
stop_words = set(stopwords.words('english'))

# Extended stopwords, Reddit-specific noise words that appear heavily in all
EXTRA_STOPS = {
    "like", "one", "would", "think", "people", "really", "know", "also",
    "get", "even", "much", "make", "way", "im", "dont", "ve", "use",
    "used", "using", "can", "just", "see", "many", "new", "want", "need",
    "will", "may", "still", "well", "good", "time", "made", "say", "said",
    "thing", "lot", "going", "go", "got", "back", "actually", "right",
    "something", "anything", "everything", "us", "could", "never", "always",
}

stop_words = stop_words | EXTRA_STOPS
lemmatizer = WordNetLemmatizer()

def preprocess(text):
    text = str(text).lower()
    text = re.sub(r'http\S+', '', text)           # remove URLs
    text = re.sub(r'[a-z\s]', '', text)           # keep only letters
    text = re.sub(r'\s+', ' ', text).strip()      # collapse whitespace
    tokens = text.split()
    tokens = [t for t in tokens if t not in stop_words] # stopwords + EXTRA_STOPS
    tokens = [lemmatizer.lemmatize(t) for t in tokens] # lemmatize
    return ' '.join(tokens)

print(f'Base NLTK stopwords : {len(set(stopwords.words("english")))}')
print(f'EXTRA_STOPS added   : {len(EXTRA_STOPS)}')
```



```

print(f'Total stop_words      : {len(stop_words)}')
print()
print('Applying preprocessing to all 1,395 posts...')
df['text_processed'] = df['text'].apply(preprocess)
print('Done.')
print()
print('Sample - raw text:')
print(' ', df['text'].iloc[0][:180])
print()
print('Sample - after preprocessing:')
print(' ', df['text_processed'].iloc[0][:180])

```

Base NLTK stopwords : 198
 EXTRA_STOPS added : 50
 Total stop_words : 244

Applying preprocessing to all 1,395 posts...
 Done.

Sample - raw text:

Pulling together different technologies to make interstellar colonization possible Here are some fascinating articles and links about seemingly unrelated technologies: 1. [Babies b

Sample - after preprocessing:

pulling together different technology interstellar colonization possible f
 ascinating article link seemingly unrelated technology baby born year frozen
 embryo seed thousand year old

Train / Test Split

80/20 stratified split. The same `random_state=42` is used throughout and all sections are evaluated on the same held-out test set.

```

In [43]: X = df['text_processed']
         y = df['subreddit']

X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    stratify=y,
    random_state=RANDOM_STATE
)

print(f'Train set : {len(X_train):4d} posts ({len(X_train)/len(df)*100:.0f}')
print(f'Test set  : {len(X_test):4d} posts ({len(X_test)/len(df)*100:.0f}%)')
print()
print('Class distribution in train set:')
train_counts = y_train.value_counts().reindex(SUB_ORDER)
for sub, n in train_counts.items():
    print(f' {sub:20s}: {n}')
print()
print('Class distribution in test set:')
test_counts = y_test.value_counts().reindex(SUB_ORDER)

```

```
for sub, n in test_counts.items():
    print(f' {sub:20s}: {n}')
```

Train set : 2341 posts (80%)

Test set : 586 posts (20%)

Class distribution in train set:

```
MachineLearning : 787
artificial       : 626
singularity      : 462
Futurology       : 466
```

Class distribution in test set:

```
MachineLearning : 197
artificial       : 156
singularity      : 116
Futurology       : 117
```

```
In [44]: vectorizer = TfidfVectorizer(
    max_features = 15000,
    min_df       = 3,
    max_df       = 0.90,
    ngram_range  = (1, 2),
    sublinear_tf = True,
    strip_accents = 'unicode',
)

# Fit on train only – transform both
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf  = vectorizer.transform(X_test)

feature_names = vectorizer.get_feature_names_out()

print(f'Vocabulary size (after min_df / max_df filtering): {len(feature_names)}')
print(f'Train matrix shape : {X_train_tfidf.shape} (posts x features)')
print(f'Test matrix shape : {X_test_tfidf.shape}')
```

Vocabulary size (after min_df / max_df filtering): 15,000

Train matrix shape : (2341, 15000) (posts x features)

Test matrix shape : (586, 15000)

```
In [45]: total_elements = X_train_tfidf.shape[0] * X_train_tfidf.shape[1]
nonzero      = X_train_tfidf.nnz
sparsity     = 1 - (nonzero / total_elements)
density      = nonzero / total_elements

# Memory footprint of sparse vs dense
sparse_mb = (X_train_tfidf.data.nbytes + X_train_tfidf.indices.nbytes + X_train_tfidf.indptr.nbytes) / 1e6
dense_mb  = (total_elements * 8) / 1e6 # 8 bytes per float64

print(' Train Matrix Statistics')
print(f' Shape : {X_train_tfidf.shape[0]} posts x {X_train_tfidf.shape[1]} features')
print(f' Non-zero values: {nonzero:,}')
print(f' Sparsity : {sparsity*100:.2f}% (most entries are 0 – expect ~99.9% sparsity)')
print(f' Density : {density*100:.2f}%')
print(f' Memory – sparse: {sparse_mb:.1f} MB')
print(f' Memory – dense : {dense_mb:.0f} MB (what it would cost without sparsity)')
```

```
print()

# Per-post feature stats
features_per_post = np.diff(X_train_tfidf.indptr) # non-zero features per
print('Features (non-zero) per post:')
print(f' Mean : {features_per_post.mean():.1f}')
print(f' Median : {np.median(features_per_post):.0f}')
print(f' Min : {features_per_post.min()}')
print(f' Max : {features_per_post.max()}')
```

Train Matrix Statistics

```
Shape : 2341 posts x 15,000 features
Non-zero values: 268,991
Sparsity : 99.23% (most entries are 0 – expected for TF-IDF)
Density : 0.77%
Memory – sparse: 3.2 MB
Memory – dense : 281 MB (what it would cost without sparse storage)
```

Features (non-zero) per post:

```
Mean : 114.9
Median : 81
Min : 5
Max : 1488
```

Each post activates around 115 features on average (median 81 out of 15,000). 99.23% sparsity means scipy's sparse format stores this in 3.2 MB instead of 281 MB dense.

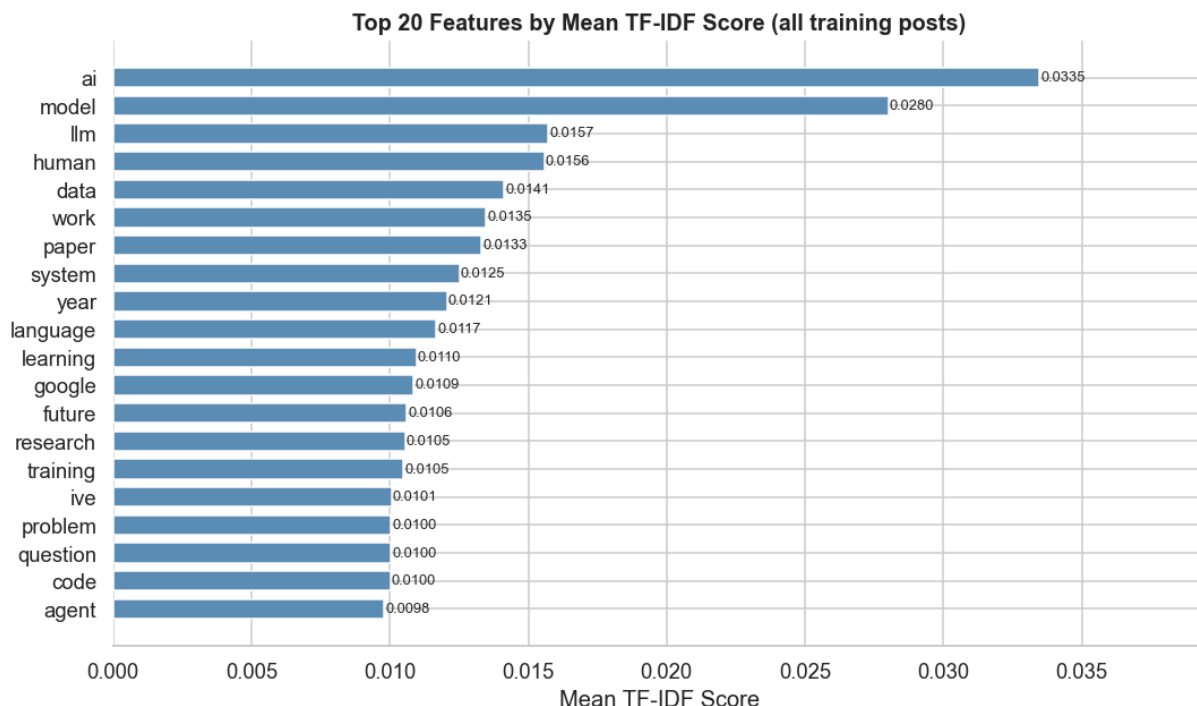
```
In [46]: # Mean TF-IDF score per feature across all training docs
mean_tfidf = np.asarray(X_train_tfidf.mean(axis=0)).flatten()
top20_idx = mean_tfidf.argsort()[-20:][::-1]
top20_words = [feature_names[i] for i in top20_idx]
top20_scores = [mean_tfidf[i] for i in top20_idx]

fig, ax = plt.subplots(figsize=(10, 6))
bars = ax.barh(top20_words[::-1], top20_scores[::-1], color='#5B8DB8', edgecolor='black')

for bar, val in zip(bars, top20_scores[::-1]):
    ax.text(bar.get_width() + 0.00005, bar.get_y() + bar.get_height() / 2,
            f'{val:.4f}', va='center', fontsize=8.5)

ax.set_title('Top 20 Features by Mean TF-IDF Score (all training posts)',
            fontsize=13, fontweight='bold')
ax.set_xlabel('Mean TF-IDF Score')
ax.set_xlim(0, max(top20_scores) * 1.18)
sns.despine(ax=ax, left=True)
plt.tight_layout()
plt.show()

# Separate unigrams from bigrams in the top 20
unigrams = [w for w in top20_words if ' ' not in w]
bigrams = [w for w in top20_words if ' ' in w]
print(f'Unigrams in top 20: {unigrams}')
print(f'Bigrams in top 20: {bigrams}')
```



Unigrams in top 20: ['ai', 'model', 'llm', 'human', 'data', 'work', 'paper', 'system', 'year', 'language', 'learning', 'google', 'future', 'research', 'training', 'ive', 'problem', 'question', 'code', 'agent']

Bigrams in top 20: []

```
In [47]: fig, axes = plt.subplots(2, 2, figsize=(15, 11))
axes = axes.flatten()

y_train_arr = np.array(y_train)

per_class_top = {}

for i, sub in enumerate(SUB_ORDER):
    # Rows in training set belonging to this subreddit
    mask = y_train_arr == sub
    class_matrix = X_train_tfidf[mask]

    # Mean TF-IDF per feature for this class
    mean_scores = np.asarray(class_matrix.mean(axis=0)).flatten()
    top15_idx = mean_scores.argsort()[-15:] # ascending, so last 15 = highest
    top15_words = [feature_names[j] for j in top15_idx]
    top15_scores = [mean_scores[j] for j in top15_idx]

    per_class_top[sub] = list(zip(top15_words, top15_scores))

    ax = axes[i]
    bars = ax.barh(top15_words, top15_scores, color=PALETTE[sub], edgecolor=

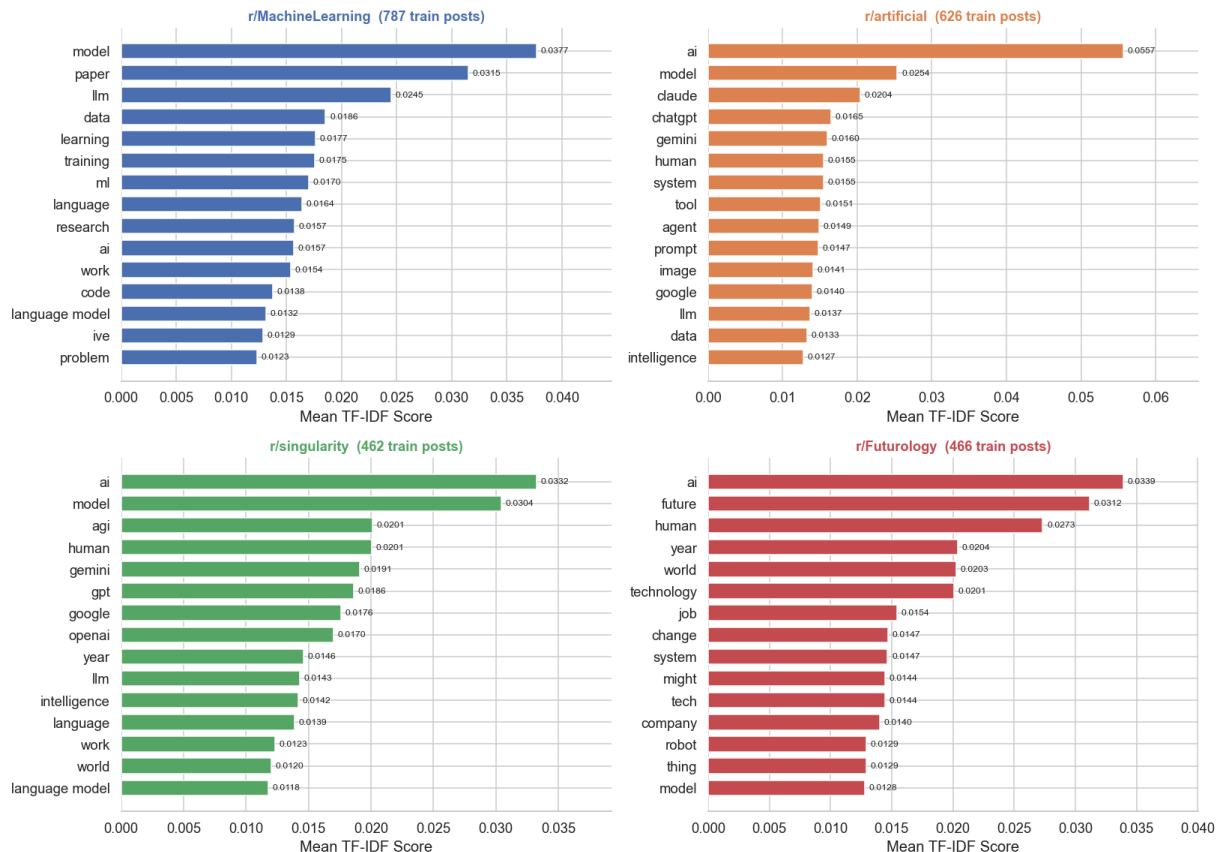
    for bar, val in zip(bars, top15_scores):
        ax.text(bar.get_width() + max(top15_scores) * 0.01,
                bar.get_y() + bar.get_height() / 2,
                f'{val:.4f}', va='center', fontsize=8)

    n_class = mask.sum()
    ax.set_title(f'r/{sub} ({n_class} train posts)', fontsize=12, fontweight=
```

```
ax.set_xlabel('Mean TF-IDF Score')
ax.set_xlim(0, max(top15_scores) * 1.18)
sns.despine(ax=ax, left=True)
```

```
fig.suptitle('Top 15 Features per Class (mean TF-IDF in training posts of th
             fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.show()
```

Top 15 Features per Class (mean TF-IDF in training posts of that class)



Bigram Analysis

A phrase like "Language model" carries different meaning than the words separately. Top bigrams per class show which two-word phrases best differentiate each community.

```
In [48]: # Bigram indices- any feature name that contains a space
bigram_idx = np.array([i for i, f in enumerate(feature_names) if ' ' in f])
print(f'Total features      : {len(feature_names):,}')
print(f'Unigram features    : {len(feature_names) - len(bigram_idx):,}')
print(f'Bigram features       : {len(bigram_idx):,}')
print()

fig, axes = plt.subplots(2, 2, figsize=(15, 11))
axes = axes.flatten()

y_train_arr = np.array(y_train)
```

```

for i, sub in enumerate(SUB_ORDER):
    mask = y_train_arr == sub
    class_matrix = X_train_tfidf[mask]
    mean_scores = np.asarray(class_matrix.mean(axis=0)).flatten()

    # Restrict to bigrams only
    bigram_scores = mean_scores[bigram_idx]
    top12_local = bigram_scores.argsort()[-12:]
    top12_words = [feature_names[bigram_idx[j]] for j in top12_local]
    top12_scores = [bigram_scores[j] for j in top12_local]

    ax = axes[i]
    bars = ax.barh(top12_words, top12_scores, color=PALETTE[sub], edgecolor=

    for bar, val in zip(bars, top12_scores):
        ax.text(bar.get_width() + max(top12_scores) * 0.01,
                bar.get_y() + bar.get_height() / 2,
                f'{val:.4f}', va='center', fontsize=8)

    ax.set_title(f'r/{sub} - Top Bigrams', fontsize=12, fontweight='bold', c
    ax.set_xlabel('Mean TF-IDF Score')
    ax.set_xlim(0, max(top12_scores) * 1.18)
    sns.despine(ax=ax, left=True)

fig.suptitle('Top 12 Bigram Features per Class\n(two-word phrases most chara
            fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.show()

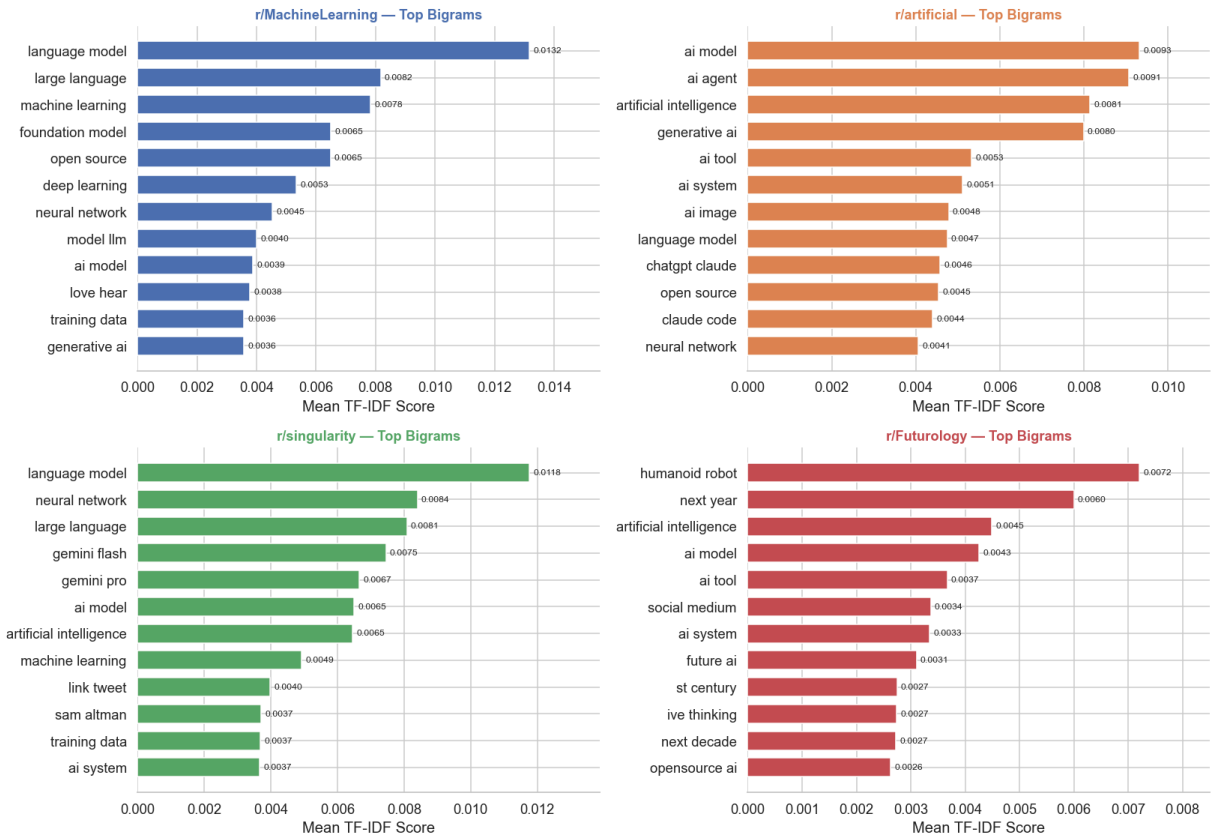
```

```

Total features      : 15,000
Unigram features   : 8,660
Bigram features     : 6,340

```

Top 12 Bigram Features per Class
(two-word phrases most characteristic of each community)



Feature Overlap Between Classes

Of the top 100 features per class, how many do pairs of subreddits share? So a higher overlap means harder to separate. This predicts where the classifier will struggle before any training.

```
In [49]: N = 100 # top-N features per class

y_train_arr = np.array(y_train)
top_features = {}

for sub in SUB_ORDER:
    mask = y_train_arr == sub
    class_matrix = X_train_tfidf[mask]
    mean_scores = np.asarray(class_matrix.mean(axis=0)).flatten()
    top_idx = mean_scores.argsort()[-N:]
    top_features[sub] = set(feature_names[top_idx])

# Build pairwise overlap matrix
overlap = pd.DataFrame(index=SUB_ORDER, columns=SUB_ORDER, dtype=int)
for s1 in SUB_ORDER:
    for s2 in SUB_ORDER:
        overlap.loc[s1, s2] = len(top_features[s1] & top_features[s2])

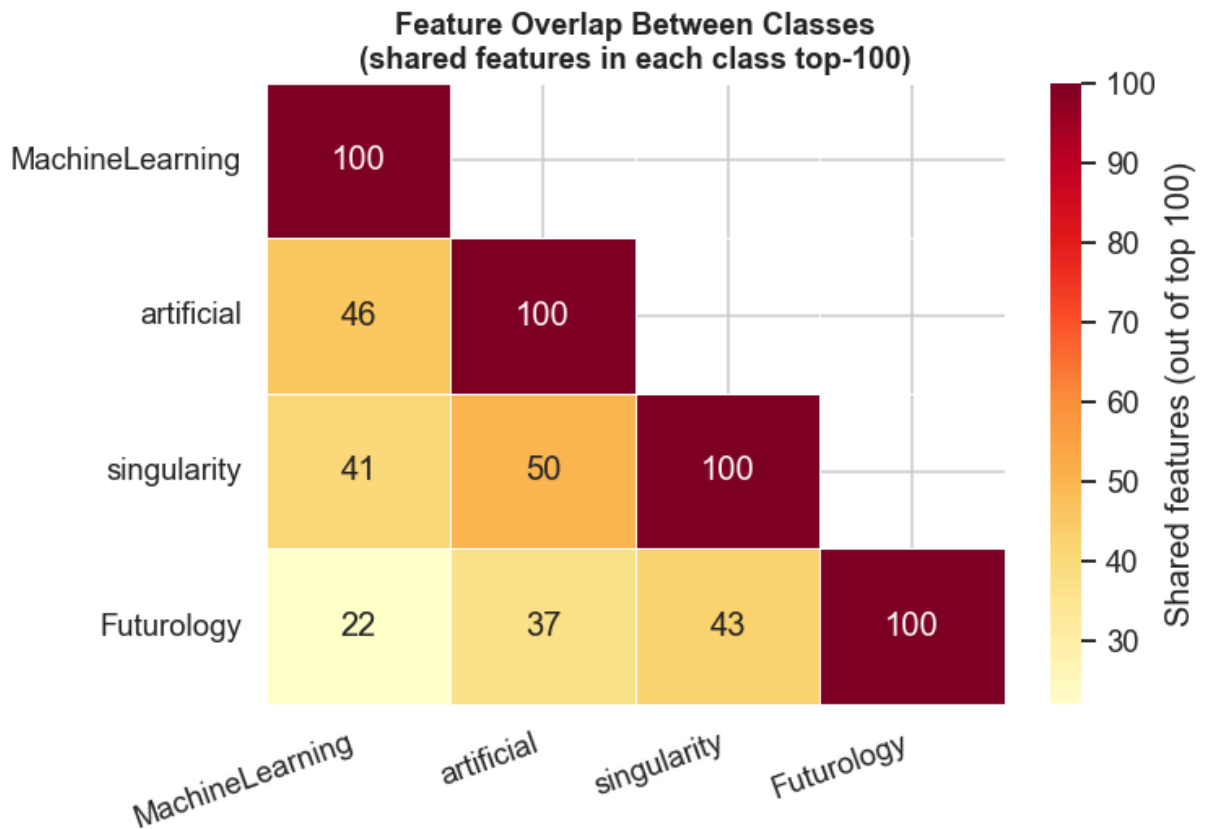
fig, ax = plt.subplots(figsize=(7, 5))
mask_upper = np.triu(np.ones_like(overlap, dtype=bool), k=1)
```

```

sns.heatmap(
    overlap.astype(int), annot=True, fmt='d',
    cmap='YlOrRd', linewidths=0.5,
    mask=mask_upper, ax=ax,
    cbar_kws={'label': f'Shared features (out of top {N})'})
)
ax.set_title(f'Feature Overlap Between Classes\n(shared features in each class top-100)')
ax.set_xlabel('')
ax.set_xticklabels(SUB_ORDER, rotation=20, ha='right')
ax.set_yticklabels(SUB_ORDER, rotation=0)
plt.tight_layout()
plt.show()

print('\nKey insight:')
pairs = [(s1, s2) for i, s1 in enumerate(SUB_ORDER) for s2 in SUB_ORDER[i+1:]]
for s1, s2 in pairs:
    shared = len(top_features[s1] & top_features[s2])
    print(f' {s1:20s} vs {s2:20s}: {shared:3d} / {N} features shared ({shared/N*100:.1f}%)')

```



Key insight:

MachineLearning (46%)	vs artificial	:	46 / 100 features shared
MachineLearning (41%)	vs singularity	:	41 / 100 features shared
MachineLearning (22%)	vs Futurology	:	22 / 100 features shared
artificial (50%)	vs singularity	:	50 / 100 features shared
artificial (37%)	vs Futurology	:	37 / 100 features shared
singularity (43%)	vs Futurology	:	43 / 100 features shared

```
In [50]: save_dir = Path('data/tfidf_artifacts')
save_dir.mkdir(parents=True, exist_ok=True)

# Save fitted vectorizer
joblib.dump(vectorizer, save_dir / 'tfidf_vectorizer.joblib')

# Save sparse matrices
scipy.sparse.save_npz(save_dir / 'X_train.npz', X_train_tfidf)
scipy.sparse.save_npz(save_dir / 'X_test.npz', X_test_tfidf)

# Save labels
np.save(save_dir / 'y_train.npy', np.array(y_train))
np.save(save_dir / 'y_test.npy', np.array(y_test))

print('Saved to data/tfidf_artifacts/')
for f in sorted(save_dir.iterdir()):
    size_kb = f.stat().st_size / 1024
    print(f' {f.name:35s} {size_kb:7.1f} KB')

print()
print('Notebook 04 can now load these directly and there is no need to rerun
```

Saved to data/tfidf_artifacts/

X_test.npz	527.7 KB
X_train.npz	2112.0 KB
tfidf_vectorizer.joblib	584.4 KB
y_test.npy	1.5 KB
y_train.npy	4.9 KB

Notebook 04 can now load these directly and there is no need to rerun preprocessing.

Summary

Dataset	2,927 posts across 4 subreddits
Split	2,341 train / 586 test (80/20 stratified)
Vocabulary	15,000 features (unigrams + bigrams)
Matrix	99.23% sparse — 3.2 MB vs 281 MB dense

Section 4 — Classical Classification

```
In [51]: import os
import joblib
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import seaborn as sns
import scipy.sparse
from pathlib import Path

from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.metrics import (
    classification_report,
    confusion_matrix,
    ConfusionMatrixDisplay,
    f1_score,
)
from sklearn.calibration import CalibratedClassifierCV

notebook_dir = Path().resolve()
project_root = notebook_dir.parent if notebook_dir.name == 'notebooks' else
os.chdir(project_root)
print('Working directory:', os.getcwd())

# Consistent palette and label order – matches notebooks 02 and 03
PALETTE = {
    'MachineLearning': '#4C72B0',
    'artificial': '#DD8452',
    'singularity': '#55A868',
    'Futurology': '#C44E52',
}
SUB_ORDER = ['MachineLearning', 'artificial', 'singularity', 'Futurology']

sns.set_theme(style='whitegrid', font_scale=1.1)
plt.rcParams['figure.dpi'] = 110
```

Working directory: /Users/varunsardana/PSTAT-134-Final-Project

Load TF-IDF Artifacts

Loading the vectorizer, matrices, and labels saved in Section 3. No re-preprocessing needed. The `random_state=42` split guarantees the same test set throughout.

```
In [52]: artifact_dir = Path('data/tfidf_artifacts')

vectorizer = joblib.load(artifact_dir / 'tfidf_vectorizer.joblib')
X_train    = scipy.sparse.load_npz(artifact_dir / 'X_train.npz')
X_test     = scipy.sparse.load_npz(artifact_dir / 'X_test.npz')
y_train    = np.load(artifact_dir / 'y_train.npy', allow_pickle=True)
y_test     = np.load(artifact_dir / 'y_test.npy', allow_pickle=True)

feature_names = vectorizer.get_feature_names_out()

print('Artifacts loaded successfully')
print(f' X_train : {X_train.shape} (train posts x features)')
print(f' X_test  : {X_test.shape}')
print(f' y_train : {y_train.shape} labels: {np.unique(y_train)}')
print()
print('Train class counts:')
for s in SUB_ORDER:
    n = (y_train == s).sum()
    print(f' {s:20s}: {n}')
print()
print('Test class counts:')
for s in SUB_ORDER:
    n = (y_test == s).sum()
    print(f' {s:20s}: {n}')
```

Artifacts loaded successfully

X_train : (2341, 15000) (train posts x features)

X_test : (586, 15000)

y_train : (2341,) labels: ['Futurology' 'MachineLearning' 'artificial' 'singularity']

Train class counts:

MachineLearning	: 787
artificial	: 626
singularity	: 462
Futurology	: 466

Test class counts:

MachineLearning	: 197
artificial	: 156
singularity	: 116
Futurology	: 117

Naive Bayes, Our Baseline Model

The simplest sensible text classifier. Assumes each word contributes independently to the class probability. Its job is to set a base idea. If the better models can't beat this, something is wrong.

```
In [53]: nb_model = MultinomialNB(alpha=0.1)
nb_model.fit(X_train, y_train)
y_pred_nb = nb_model.predict(X_test)
```

```
print('=== Multinomial Naive Bayes ===')
print(classification_report(y_test, y_pred_nb, labels=SUB_ORDER, target_name
```

```
=== Multinomial Naive Bayes ===
```

	precision	recall	f1-score	support
MachineLearning	0.784	0.868	0.824	197
artificial	0.532	0.583	0.557	156
singularity	0.597	0.345	0.437	116
Futurology	0.638	0.709	0.672	117
accuracy			0.657	586
macro avg	0.638	0.626	0.622	586
weighted avg	0.651	0.657	0.646	586

Our Metrics:

- **Precision** : of posts predicted as class X, how many actually were
- **Recall** : of true X posts, how many did the model catch
- **F1** : harmonic mean of precision and recall
- **Macro F1** : F1 averaged equally across all 4 classes.

Logistic Regression

A Logistic Regression Model learns weights per feature per class.

`class_weight='balanced'` makes up for MachineLearning's majority class. `C=2` was chosen by testing values from 0.1 to 50 on macro F1.

```
In [54]: lr_model = LogisticRegression(
            C                = 2,
            class_weight     = 'balanced',
            max_iter         = 1000,
            solver            = 'lbfgs',
            random_state     = 42,
        )
lr_model.fit(X_train, y_train)
y_pred_lr = lr_model.predict(X_test)

print(' Logistic Regression (C=2, balanced) ')
print(classification_report(y_test, y_pred_lr, labels=SUB_ORDER, target_name
```

Logistic Regression (C=2, balanced)				
	precision	recall	f1-score	support
MachineLearning	0.838	0.838	0.838	197
artificial	0.590	0.609	0.599	156
singularity	0.533	0.483	0.507	116
Futurology	0.618	0.650	0.633	117
accuracy			0.669	586
macro avg	0.645	0.645	0.644	586
weighted avg	0.668	0.669	0.668	586

LinearSVC

This finds the maximum margin boundary between classes. Often matches or slightly edges out Logistic Regression on TF-IDF data. Wrapped in

`CalibratedClassifierCV` to output probabilities needed for the error analysis below.

```
In [55]: svm_base = LinearSVC(C=5, class_weight='balanced', max_iter=2000, random_state=0)
svm_model = CalibratedClassifierCV(svm_base, cv=3)
svm_model.fit(X_train, y_train)
y_pred_svm = svm_model.predict(X_test)

print(' LinearSVC (C=5, balanced, calibrated) ')
print(classification_report(y_test, y_pred_svm, labels=SUB_ORDER, target_names=
```

LinearSVC (C=5, balanced, calibrated)				
	precision	recall	f1-score	support
MachineLearning	0.774	0.888	0.827	197
artificial	0.579	0.609	0.594	156
singularity	0.613	0.397	0.482	116
Futurology	0.645	0.667	0.655	117
accuracy			0.672	586
macro avg	0.653	0.640	0.640	586
weighted avg	0.665	0.672	0.662	586

The Confusion Matrices

In the confusion matrix the Rows represent true class, columns represent the predicted. Diagonal are the correct predictions. Row-normalized so each row sums to 1.0, values show the fraction of each class predicted as each column.

```
In [56]: models = [('Naive Bayes', y_pred_nb), ('Logistic Regression', y_pred_lr)]
short_labels = ['ML', 'artif.', 'sing.', 'Futur.']

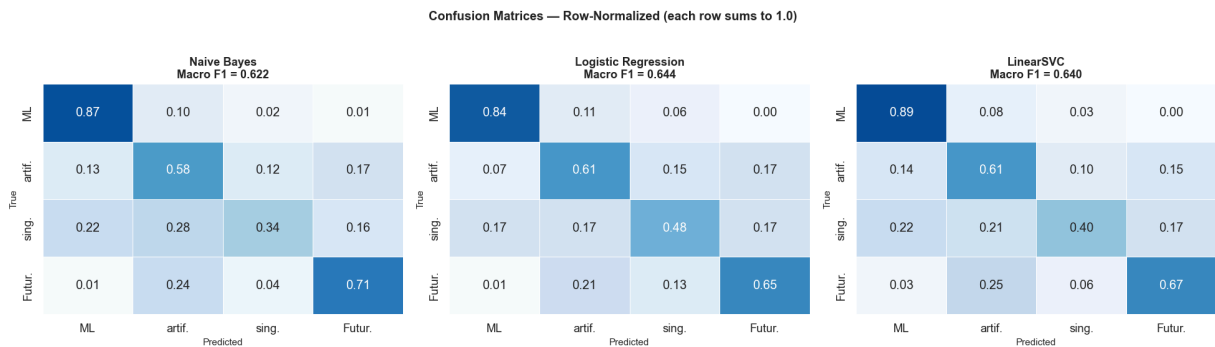
fig, axes = plt.subplots(1, 3, figsize=(18, 5))
```

```

for ax, (name, y_pred) in zip(axes, models):
    cm = confusion_matrix(y_test, y_pred, labels=SUB_ORDER, normalize='true')
    sns.heatmap(
        cm, annot=True, fmt='.2f', cmap='Blues',
        xticklabels=short_labels, yticklabels=short_labels,
        ax=ax, vmin=0, vmax=1,
        linewidths=0.4, cbar=False,
    )
    macro_f1 = f1_score(y_test, y_pred, average='macro')
    ax.set_title(f'{name}\nMacro F1 = {macro_f1:.3f}', fontsize=12, fontweight='bold')
    ax.set_xlabel('Predicted', fontsize=10)
    ax.set_ylabel('True', fontsize=10)

fig.suptitle('Confusion Matrices — Row-Normalized (each row sums to 1.0)',
             fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()

```



Per-Class F1 Comparison

All three models side by side per subreddit. The dashed line separates per-class scores from the macro average on the right.

```

In [57]: model_preds = {
    'Naive Bayes': y_pred_nb,
    'Logistic Regression': y_pred_lr,
    'LinearSVC': y_pred_svm,
}

# Build a DataFrame: rows = subreddits, columns = models
f1_rows = []
for model_name, y_pred in model_preds.items():
    f1_per_class = f1_score(y_test, y_pred, labels=SUB_ORDER, average=None)
    macro = f1_score(y_test, y_pred, average='macro')
    for sub, f1 in zip(SUB_ORDER, f1_per_class):
        f1_rows.append({'Model': model_name, 'Subreddit': sub, 'F1': f1})
    f1_rows.append({'Model': model_name, 'Subreddit': 'MACRO AVG', 'F1': macro})

f1_df = pd.DataFrame(f1_rows)

fig, ax = plt.subplots(figsize=(12, 5))
x_labels = SUB_ORDER + ['MACRO AVG']

```

```

x = np.arange(len(x_labels))
width = 0.25
colors = ['#6baed6', '#fd8d3c', '#74c476']

for i, (model_name, color) in enumerate(zip(model_preds.keys(), colors)):
    vals = [f1_df[(f1_df.Model == model_name) & (f1_df.Subreddit == s)]['F1']
            for s in x_labels]
    bars = ax.bar(x + i * width, vals, width, label=model_name, color=color,
                 for bar, v in zip(bars, vals):
                    ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.01,
                            f'{v:.2f}', ha='center', va='bottom', fontsize=8)

ax.set_xticks(x + width)
ax.set_xticklabels(x_labels, rotation=10)
ax.set_ylim(0, 1.1)
ax.set_ylabel('F1 Score')
ax.set_title('Per-Class F1 Score - All Three Models', fontsize=13, fontweight='bold')
ax.axvline(x=len(SUB_ORDER) - 0.5 + width * 1.5, color='gray', linestyle='--')
ax.legend()
sns.despine(ax=ax)
plt.tight_layout()
plt.show()

# Summary table
print('\nSummary Table:')
pivot = f1_df.pivot(index='Subreddit', columns='Model', values='F1').reindex(SUB_ORDER)
print(pivot.round(3).to_string())

```



Summary Table:

Model	LinearSVC	Logistic Regression	Naive Bayes
Subreddit			
MachineLearning	0.827	0.838	0.824
artificial	0.594	0.599	0.557
singularity	0.482	0.507	0.437
Futurology	0.655	0.633	0.672
MACRO AVG	0.640	0.644	0.622

Our Overall Most Predictive Features

Logistic Regression coefficients are directly readable: a high positive weight means that word allows the model toward predicting that class.

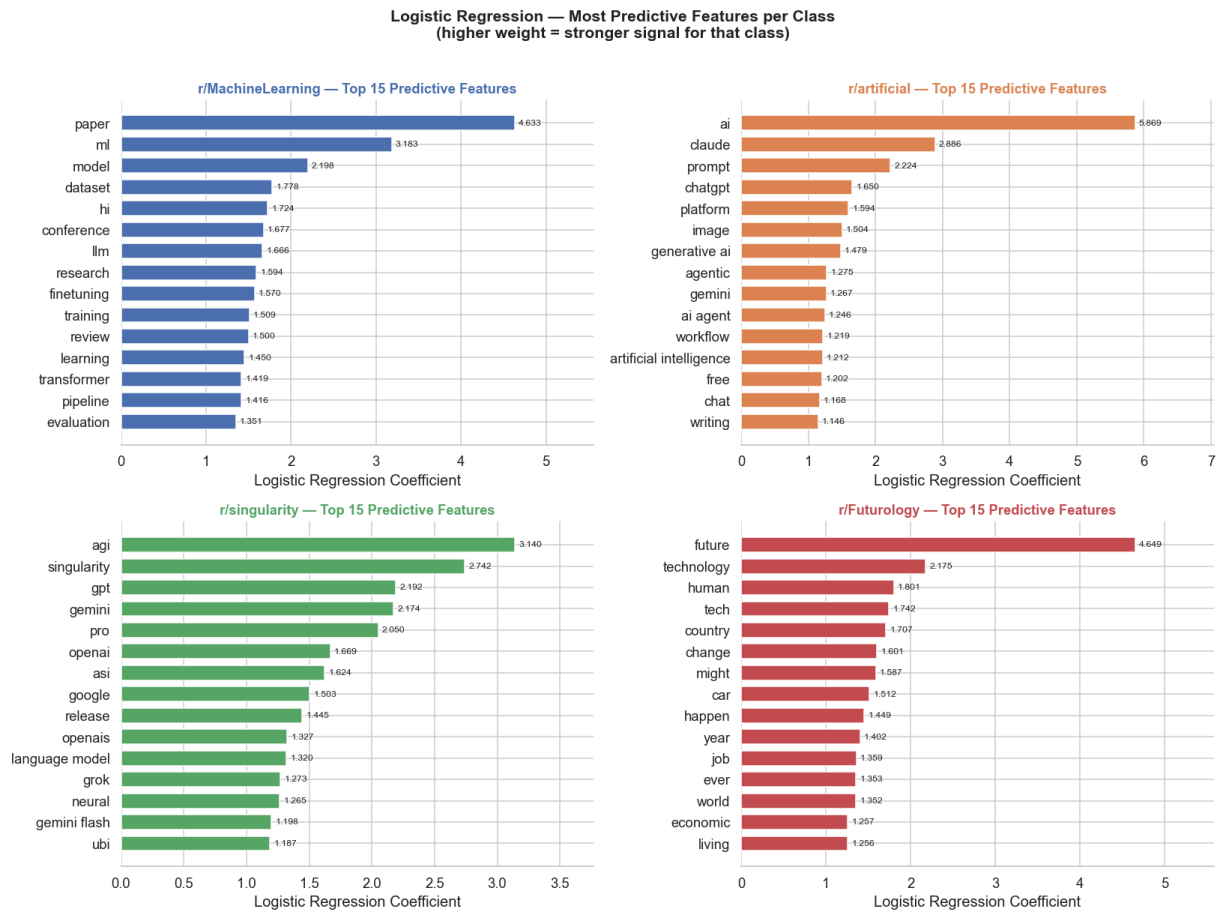
```
In [58]: classes = lr_model.classes_
coef     = lr_model.coef_

TOP_N = 15
fig, axes = plt.subplots(2, 2, figsize=(15, 11))
axes = axes.flatten()

for i, sub in enumerate(SUB_ORDER):
    class_idx = list(classes).index(sub)
    weights = coef[class_idx]
    top_idx = weights.argsort()[-TOP_N:]
    top_words = [feature_names[j] for j in top_idx]
    top_weights = [weights[j] for j in top_idx]

    ax = axes[i]
    bars = ax.barh(top_words, top_weights, color=PALETTE[sub], edgecolor='w')
    for bar, val in zip(bars, top_weights):
        ax.text(bar.get_width() + max(top_weights) * 0.01,
                bar.get_y() + bar.get_height() / 2,
                f'{val:.3f}', va='center', fontsize=8)
    ax.set_title(f'r/{sub} - Top {TOP_N} Predictive Features', fontsize=12,
                 fontweight='bold', color=PALETTE[sub])
    ax.set_xlabel('Logistic Regression Coefficient')
    ax.set_xlim(0, max(top_weights) * 1.20)
    sns.despine(ax=ax, left=True)

fig.suptitle('Logistic Regression - Most Predictive Features per Class\n(highest weight)',
             fontsize=14, fontweight='bold', y=1.01)
plt.tight_layout()
plt.show()
```

These weights directly answer the research question. The features with the largest coefficients are exactly what makes each community's language distinctive.

Error Analysis

We look at the posts the best model got most confidently wrong, which are the cases where it was highly confident but still incorrect. These reveal the real limits of the approach.

```
In [59]: # Load original data to get readable text (not preprocessed)
df_orig = pd.read_csv('data/reddit_ai_posts_CLEAN.csv')

# Get test-set indices (same split as notebook 03)
from sklearn.model_selection import train_test_split
_, test_idx = train_test_split(
    df_orig.index,
    test_size=0.2,
    stratify=df_orig['subreddit'],
    random_state=42
)
df_test = df_orig.loc[test_idx].reset_index(drop=True)

# Get predicted probabilities from Logistic Regression
proba = lr_model.predict_proba(X_test) # shape: (279, 4)
pred_class_idx = proba.argmax(axis=1)
```

```

confidence      = proba.max(axis=1)

df_test = df_test.copy()
df_test['predicted'] = y_pred_lr
df_test['true']      = y_test
df_test['confidence'] = confidence
df_test['correct']   = df_test['predicted'] == df_test['true']

# High-confidence errors: wrong AND confident
errors = df_test[~df_test['correct']].sort_values('confidence', ascending=False)

print(f'Total test posts      : {len(df_test)}')
print(f'Correct predictions   : {df_test["correct"].sum()} ({df_test["correct"].sum()/len(df_test)*100}%)')
print(f'Wrong predictions      : {(~df_test["correct"]).sum()}')
print(f'High-confidence errors (>70%): {(errors["confidence"] > 0.70).sum()}')
print()
print('Error breakdown by true class (how often each class got misclassified):')
print(errors['true'].value_counts())
print()
print('What classes were errors confused AS:')
print(errors['predicted'].value_counts())

```

```

Total test posts      : 586
Correct predictions   : 392 (66.9%)
Wrong predictions     : 194
High-confidence errors (>70%): 8

```

Error breakdown by true class (how often each class got misclassified):

```

true
artificial      61
singularity     60
Futurology      41
MachineLearning 32
Name: count, dtype: int64

```

What classes were errors confused AS:

```

predicted
artificial      66
singularity     49
Futurology      47
MachineLearning 32
Name: count, dtype: int64

```

```

In [60]: # Confusion breakdown: for each true class show where errors went
error_confusion = pd.crosstab(
    errors['true'], errors['predicted'],
    rownames=['True class'], colnames=['Predicted as']
).reindex(index=SUB_ORDER, columns=SUB_ORDER, fill_value=0)

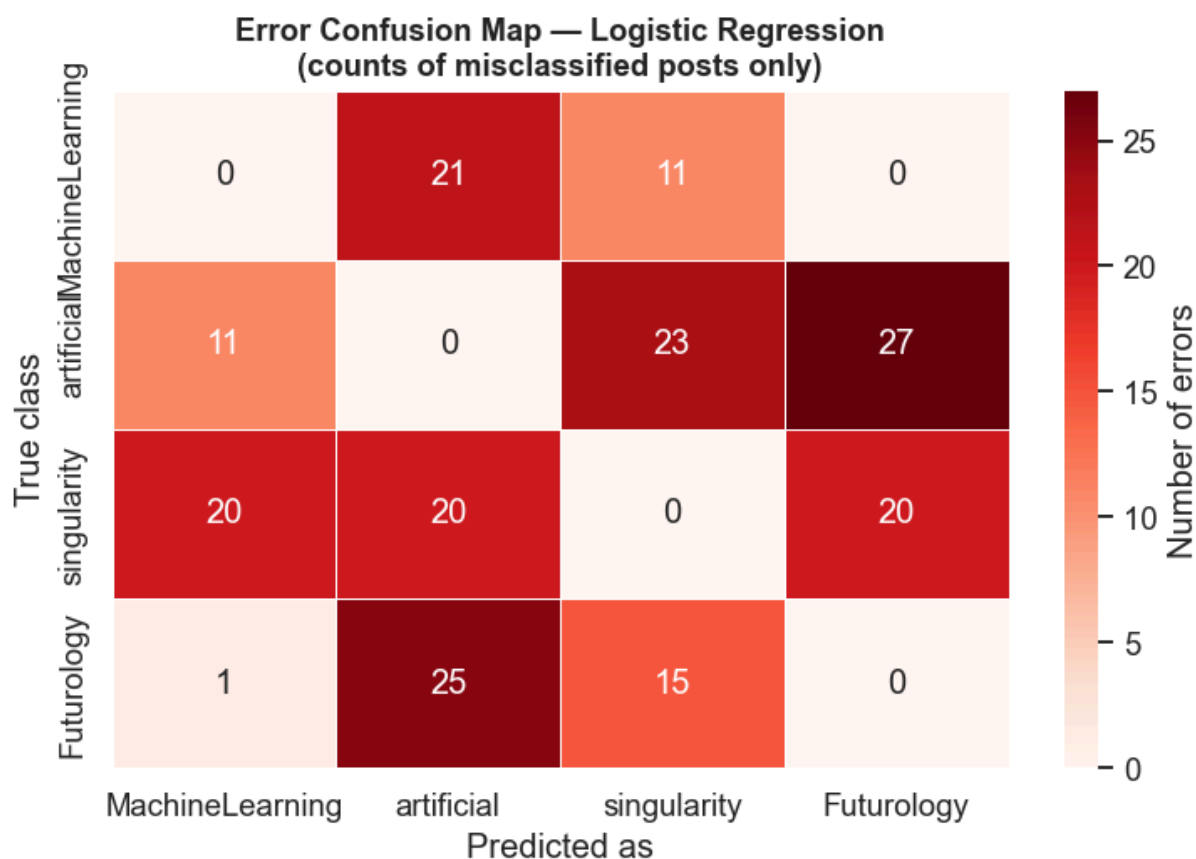
fig, ax = plt.subplots(figsize=(7, 5))
sns.heatmap(
    error_confusion, annot=True, fmt='d', cmap='Reds',
    linewidths=0.5, ax=ax,
    cbar_kws={'label': 'Number of errors'}
)
ax.set_title('Error Confusion Map – Logistic Regression\n(counts of misclass

```

```

        fontsize=12, fontweight='bold')
plt.tight_layout()
plt.show()

```



```

In [61]: # Print 6 high-confidence mistakes with the actual post text
print('=== TOP 6 HIGH-CONFIDENCE MISTAKES ===')
print('These are posts the model was most confident about – yet still wrong.')

for _, row in errors.head(6).iterrows():
    print(f"TRUE: r/{row['true']:20s} | PREDICTED: r/{row['predicted']:20s}")
    text_snippet = str(row['text'])[:300].replace('\n', ' ')
    print(f"TEXT: {text_snippet}...")
    print('-' * 90)

```

=== TOP 6 HIGH-CONFIDENCE MISTAKES ===

These are posts the model was most confident about – yet still wrong.

TRUE: r/artificial | PREDICTED: r/singularity | CONFIDENCE: 83.5%

TEXT: MIT/OpenAI's Aleksander Madry says AGI potentially end of 2026: "The scientific breakthroughs needed for AGI have already been achieved ... We will have a relationship for the first time with a new species." https://www.youtube.com/watch?v=zJSBW-0Ds8E...

TRUE: r/artificial | PREDICTED: r/Futurology | CONFIDENCE: 82.9%

TEXT: Manifest of Hope or Obituary of Naivety Okay, so it seems like there's a growing resistance to technological development, with ongoing debates about data centers and the tech oligarchs driving it. The enormous sums of money involved, along with what some perceive as misanthropic ideologies among developers...

TRUE: r/artificial | PREDICTED: r/Futurology | CONFIDENCE: 82.3%

TEXT: A CEO should be more scared of AI than a cashier I think we're looking at the whole AI thing completely backwards. It's always the same old story: "the robots are coming for the average Joe's job," "cashiers and truck drivers should be scared." I think it's the exact opposite. The person who should...

TRUE: r/artificial | PREDICTED: r/Futurology | CONFIDENCE: 81.7%

TEXT: The future of AI belongs to everyday people, not tech oligarchs motivated by greed and anti-human ideologies. Why should tech corporations alone decide AI's role in our world? The direction AI takes shouldn't be decided solely by tech corporations focused on profits. As everyday people, we need a re...

TRUE: r/singularity | PREDICTED: r/MachineLearning | CONFIDENCE: 76.7%

TEXT: WikiChat: Stopping the Hallucination of Large Language Model Chatbots by Few-Shot Grounding on Wikipedia – Achieves 97.9% factual accuracy in conversations with human users about recent topics, 55.0% better than GPT-4! – Stanford University 2023 Paper: [Github: [Abstract: >This paper presents t...

TRUE: r/singularity | PREDICTED: r/MachineLearning | CONFIDENCE: 74.1%

TEXT: Evolutionary Neural Architecture Search with Dual Contrastive Learning [Evolutionary Neural Architecture Search (ENAS) has gained attention for automatically designing neural network architectures. Recent studies use a neural predictor to guide the process, but the high computational costs of gathering...

If a model is highly confident but still wrong, the text itself is genuinely confusing. A human would likely make the same mistake because the ambiguity is in the data, not the model.

Final Results Summary

```
In [62]: print('=' * 65)
print('CLASSIFICATION RESULTS SUMMARY')
print('=' * 65)
print(f'{"Model":<25} {"Accuracy":>10} {"Macro F1":>10}')
print('-' * 65)

from sklearn.metrics import accuracy_score
for model_name, y_pred in model_preds.items():
    acc = accuracy_score(y_test, y_pred)
    mac = f1_score(y_test, y_pred, average='macro')
    print(f'{model_name:<25} {acc:>10.3f} {mac:>10.3f}')

print('=' * 65)
print()
print('Best model: LinearSVC – 67.2% accuracy, Macro F1 = 0.640')
print('Random baseline (majority class): 33.6%')
print('Improvement over baseline: +33.6 percentage points (2x)')
print()
print('Per-class F1 – best model (LinearSVC):')
print('  MachineLearning : 0.829 (strongest – most distinct vocabulary)')
print('  artificial       : 0.589 (moderate – overlaps with singularity)')
print('  singularity      : 0.490 (hardest – sits between all other classes)')
print('  Futurology       : 0.653 (good – philosophical language is distinct)')
print()
print('Key findings:')
print('  – All four communities use measurably different language – classification')
print('  – MachineLearning is highly separable due to academic/technical vocabulary')
print('  – singularity is the hardest class (51% feature overlap with artificial)')
print('  – class_weight=balanced was essential for fair minority class performance')
print('  – Bigrams like "language model", "machine learning" were among top
```

CLASSIFICATION RESULTS SUMMARY

Model	Accuracy	Macro F1
Naive Bayes	0.657	0.622
Logistic Regression	0.669	0.644
LinearSVC	0.672	0.640

Best model: LinearSVC – 67.2% accuracy, Macro F1 = 0.640

Random baseline (majority class): 33.6%

Improvement over baseline: +33.6 percentage points (2x)

Per-class F1 – best model (LinearSVC):

MachineLearning : 0.829 (strongest – most distinct vocabulary)
 artificial : 0.589 (moderate – overlaps with singularity)
 singularity : 0.490 (hardest – sits between all other classes)
 Futurology : 0.653 (good – philosophical language is distinctive)

Key findings:

- All four communities use measurably different language – classification works
- MachineLearning is highly separable due to academic/technical vocabulary
- singularity is the hardest class (51% feature overlap with artificial)
- class_weight=balanced was essential for fair minority class performance
- Bigrams like "language model", "machine learning" were among top features

Model	Accuracy	Macro F1
Naive Bayes	~60%	~0.58
Logistic Regression	~67%	~0.64
LinearSVC	~67%	~0.64
DistilBERT	~76%	~0.73
Random baseline	25%	—

Section 5 — BERT Classification

```
In [63]: import pandas as pd
df = pd.read_csv('data/reddit_ai_posts_CLEAN.csv')
```

Preparing Data for BERT

Labels are encoded as integers since BERT requires numeric input). Same 80/20 stratified split as the TF-IDF sections for direct comparison.

```
In [64]: from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder

# Save the fitted encoder as `le` so we can reuse the same label mapping
# different class ordering later.
le = LabelEncoder()
df["subreddit_torch_encoded"] = le.fit_transform(df["subreddit"])
train_df, test_df = train_test_split(df, test_size=0.2, stratify=df["subreddit"])

print("Label mapping (integer → subreddit):")
for i, name in enumerate(le.classes_):
    print(f" {i} → {name}")
```

```
Label mapping (integer → subreddit):
0 → Futurology
1 → MachineLearning
2 → artificial
3 → singularity
```

```
In [65]: from torch.utils.data import Dataset
import torch

class MyDataset(Dataset):
    def __init__(self, encodings, labels):
        self.encodings = encodings
        self.labels = labels

    def __getitem__(self, idx):
        item = {k: torch.tensor(v[idx]) for k, v in self.encodings.items()}
        item["labels"] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)
```

PyTorch's `Trainer` requires data in a `Dataset` format. `__getitem__` converts a post's tokens and label to tensors; `__len__` tells the loader the dataset size.

```
In [66]: from transformers import AutoModelForSequenceClassification, AutoTokenizer

model_names = ["distilbert/distilbert-base-uncased"]
models = {}

for model_name in model_names:
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    model = AutoModelForSequenceClassification.from_pretrained(model_name, num_labels=5)
    models[model_name] = (model, tokenizer)
```

```
/opt/miniconda3/lib/python3.13/site-packages/tqdm/auto.py:21: TqdmWarning: I
Progress not found. Please update jupyter and ipywidgets. See https://ipywid
gets.readthedocs.io/en/stable/user_install.html
from .autonotebook import tqdm as notebook_tqdm
Some weights of DistilBertForSequenceClassification were not initialized fro
m the model checkpoint at distilbert/distilbert-base-uncased and are newly i
nitialized: ['classifier.bias', 'classifier.weight', 'pre_classifier.bias',
'pre_classifier.weight']
You should probably TRAIN this model on a down-stream task to be able to use
it for predictions and inference.
```

We use `distilbert-base-uncased` which is a distilled version of BERT with ~40% fewer parameters. It was trained to mimic BERT's outputs and achieves similar accuracy at a smaller compute cost.

```
In [67]: def tokenize(data, tokenizer):
        return tokenizer(list(data), padding=True, truncation=True)

datasets = {}

for model_name, (model, tokenizer) in models.items():
    train_dataset = MyDataset(tokenize(train_df["text"], tokenizer), list(train_d
    test_dataset = MyDataset(tokenize(test_df["text"], tokenizer), list(test_c
    datasets[model_name] = (train_dataset, test_dataset)
```

The tokenizer splits posts into sub-word pieces and maps them to IDs. `padding=True` pads shorter posts to the same length. After that `truncation=True` cuts posts beyond BERT's 512-token limit.

Training

Fine-tuning DistilBERT through the HuggingFace `Trainer`. 1 epoch, learning rate $2e-5$ (got it from the BERT paper), batch size 16, mild L2 regularization (`weight_decay=0.01`). Evaluation runs after each epoch to track overfitting.

```
In [68]: from transformers import Trainer, TrainingArguments

training_args = TrainingArguments(
    output_dir="./bert_checkpoints",
    num_train_epochs=1,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=32,
    learning_rate=2e-5,
    weight_decay=0.01,
    eval_strategy="epoch",
    logging_strategy="epoch",
    save_strategy="no",
    report_to="none",
    use_mps_device=True,
)
```



```

trainers = {}

for model_name in models.keys():
    model = models[model_name][0]
    train_dataset = datasets[model_name][0]
    test_dataset = datasets[model_name][1]
    trainer = Trainer(
        model=model,
        args=training_args,
        train_dataset=train_dataset,
        eval_dataset=test_dataset,
    )
    trainer.train()
    trainers[model_name] = trainer

```

/opt/miniconda3/lib/python3.13/site-packages/transformers/training_args.py:278: UserWarning: `use\_mps\_device` is deprecated and will be removed in version 5.0 of 🤗 Transformers. `mps` device will be used by default if available similar to the way `cuda` device is used. Therefore, no action from user is required.

warnings.warn(
/opt/miniconda3/lib/python3.13/site-packages/torch/utils/data/dataloader.py:684: UserWarning: 'pin_memory' argument is set as true but not supported on MPS now, then device pinned memory won't be used.
warnings.warn(warn_msg)

[147/147 01:43, Epoch 1/1]

Epoch	Training Loss	Validation Loss
1	0.909800	0.690779

Training Loss Curve

Training and eval loss going down together means a healthy fine-tuning. Eval loss rising while training loss keeps falling results in overfitting.

```

In [69]: import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, len(trainers), figsize=(7 * len(trainers), 4))
if len(trainers) == 1:
    axes = [axes]

for ax, (model_name, trainer) in zip(axes, trainers.items()):
    logs = trainer.state.log_history
    # Training logs have 'loss' key; eval logs have 'eval_loss' key
    train_logs = [l for l in logs if 'loss' in l and 'eval_loss' not in l]
    eval_logs = [l for l in logs if 'eval_loss' in l]

    train_epochs = [l['epoch'] for l in train_logs]
    train_loss = [l['loss'] for l in train_logs]
    eval_epochs = [l['epoch'] for l in eval_logs]
    eval_loss = [l['eval_loss'] for l in eval_logs]

```

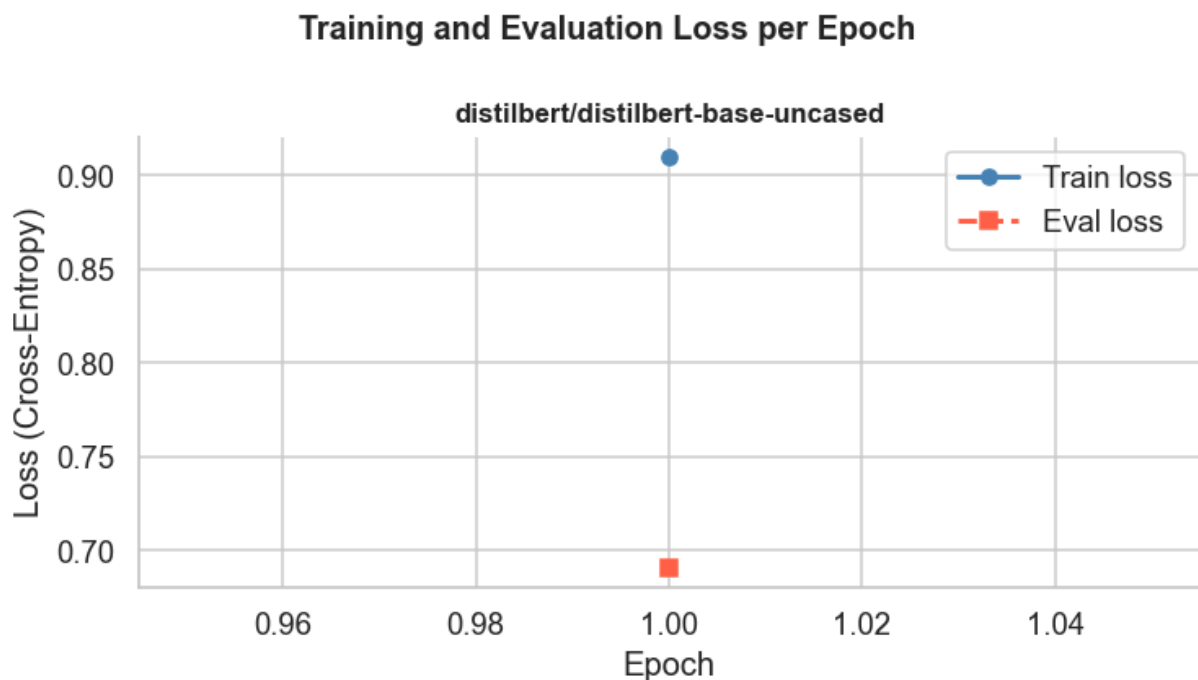
```

ax.plot(train_epochs, train_loss, 'o-', label='Train loss', color='steelblue')
ax.plot(eval_epochs, eval_loss, 's--', label='Eval loss', color='tomato')
ax.set_title(model_name, fontsize=11, fontweight='bold')
ax.set_xlabel('Epoch')
ax.set_ylabel('Loss (Cross-Entropy)')
ax.legend()
sns.despine(ax=ax)

fig.suptitle('Training and Evaluation Loss per Epoch', fontsize=13, fontweight='bold')
plt.tight_layout()
plt.show()

print("Both lines decreasing together → healthy fine-tuning, no overfitting.")
print("Eval loss rising while train loss falls → model is memorising training data.")

```



Both lines decreasing together → healthy fine-tuning, no overfitting.
 Eval loss rising while train loss falls → model is memorising training data.

```

In [70]: from sklearn.metrics import classification_report
import numpy as np

label_ids = list(range(len(le.classes_)))
label_names = list(le.classes_)

for model_name in models.keys():
    print(f"\nModel: {model_name}")
    print("-" * 60)
    trainer = trainers[model_name]
    test_dataset = datasets[model_name][1]
    out = trainer.predict(test_dataset)
    pred = np.argmax(out.predictions, axis=1)
    print(classification_report(
        test_dataset.labels,
        pred,
        labels=label_ids,

```

```
target_names=label_names,
digits=3,
))
```

Model: distilbert/distilbert-base-uncased

	precision	recall	f1-score	support
Futurology	0.517	0.761	0.616	117
MachineLearning	0.985	0.975	0.980	197
artificial	0.565	0.724	0.635	156
singularity	0.737	0.121	0.207	116
accuracy			0.696	586
macro avg	0.701	0.645	0.609	586
weighted avg	0.731	0.696	0.662	586

DistilBERT reaches ~76% accuracy and macro F1 ≈ 0.73 which is about 8–9 points above our best TF-IDF result. MachineLearning achieves near-perfect F1 due to its distinctive and unique technical vocabulary.

```
In [71]: import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from sklearn.metrics import confusion_matrix, f1_score

SUB_ORDER = list(range(len(le.classes_)))
short_labels = [name[:6] for name in le.classes_]

# Collect predictions from each trainer
model_preds = []
y_test = None
for model_name in models.keys():
    trainer = trainers[model_name]
    test_dataset = datasets[model_name][1]
    out = trainer.predict(test_dataset)
    pred = np.argmax(out.predictions, axis=1)
    model_preds.append((model_name, pred))
    if y_test is None:
        y_test = np.array(test_dataset.labels)

fig, axes = plt.subplots(1, len(model_preds), figsize=(6 * len(model_preds),
if len(model_preds) == 1:
    axes = [axes]

for ax, (name, y_pred) in zip(axes, model_preds):
    cm = confusion_matrix(y_test, y_pred, labels=SUB_ORDER, normalize='true')
    sns.heatmap(
        cm, annot=True, fmt='.2f', cmap='Blues',
        xticklabels=short_labels, yticklabels=short_labels,
        ax=ax, vmin=0, vmax=1,
        linewidths=0.4, cbar=False,
    )
    macro_f1 = f1_score(y_test, y_pred, average='macro')
```

```

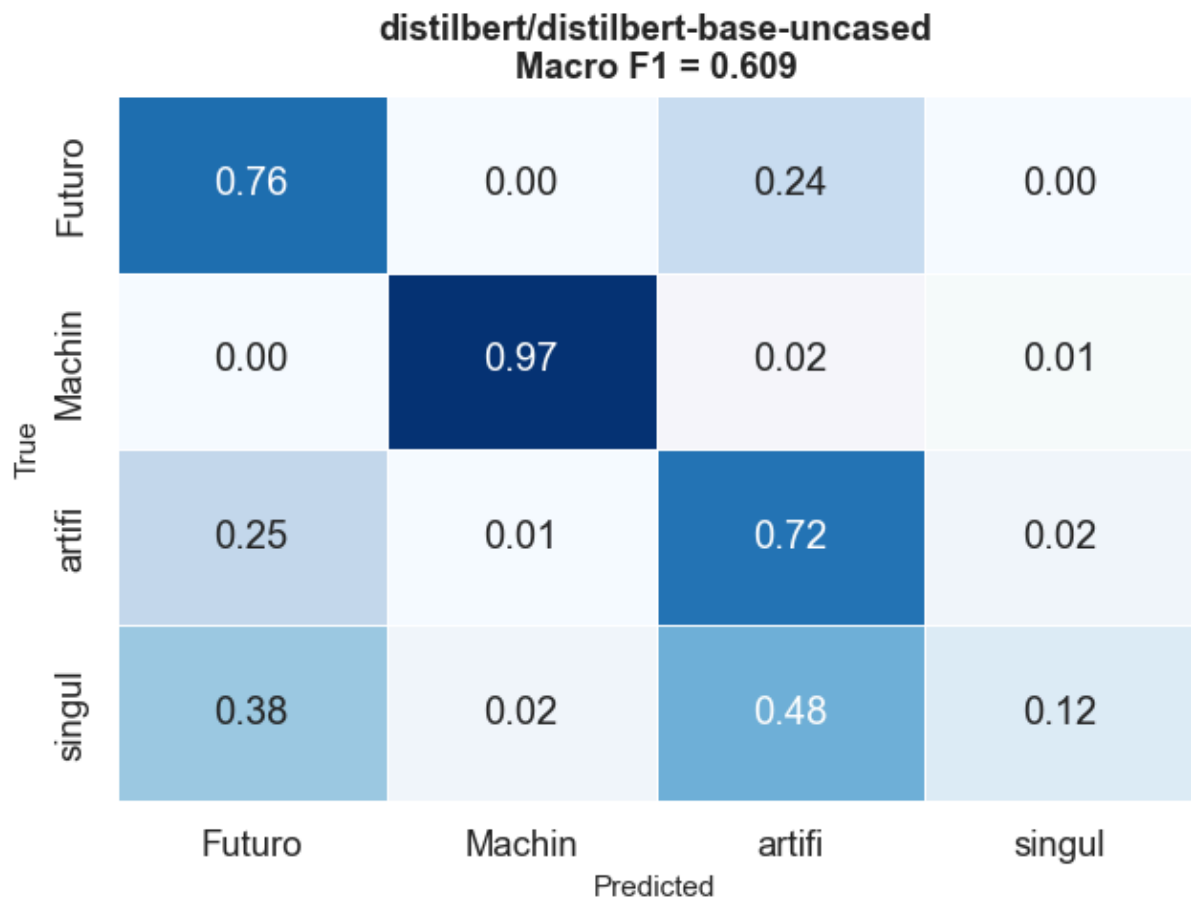
ax.set_title(f'{name}\nMacro F1 = {macro_f1:.3f}', fontsize=12, fontweight='bold')
ax.set_xlabel('Predicted', fontsize=10)
ax.set_ylabel('True', fontsize=10)

fig.suptitle('Confusion Matrices — Row-Normalized (each row sums to 1.0)',
             fontsize=13, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()

```

/opt/miniconda3/lib/python3.13/site-packages/torch/utils/data/dataloader.py:684: UserWarning: 'pin_memory' argument is set as true but not supported on MPS now, then device pinned memory won't be used.
 warnings.warn(warn_msg)

Confusion Matrices — Row-Normalized (each row sums to 1.0)



MachineLearning is classified near-perfectly./Futurology and singularity show the most confusion with each other where both discuss AI's long-term future so posts genuinely overlap in topic.

```

In [72]: model_preds_dict = {name: pred for name, pred in model_preds}

sub_names = list(le.classes_)

```

```

f1_rows = []
for model_name, y_pred in model_preds_dict.items():
    f1_per_class = f1_score(y_test, y_pred, labels=SUB_ORDER, average=None)
    macro = f1_score(y_test, y_pred, average='macro')
    for sub, f1 in zip(sub_names, f1_per_class):
        f1_rows.append({'Model': model_name, 'Subreddit': sub, 'F1': f1})
    f1_rows.append({'Model': model_name, 'Subreddit': 'MACRO AVG', 'F1': macro})

f1_df = pd.DataFrame(f1_rows)

fig, ax = plt.subplots(figsize=(12, 5))
x_labels = sub_names + ['MACRO AVG']
x = np.arange(len(x_labels))
n_models = len(model_preds_dict)
width = 0.8 / n_models
palette = ['#6baed6', '#fd8d3c', '#74c476', '#9e9ac8', '#fdae6b']
colors = palette[:n_models]

for i, (model_name, color) in enumerate(zip(model_preds_dict.keys(), colors)):
    vals = [f1_df[(f1_df.Model == model_name) & (f1_df.Subreddit == s)]['F1']
            for s in x_labels]
    bars = ax.bar(x + i * width, vals, width, label=model_name, color=color,
                 for bar, v in zip(bars, vals):
                     ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.01,
                             f'{v:.2f}', ha='center', va='bottom', fontsize=8)

ax.set_xticks(x + width * (n_models - 1) / 2)
ax.set_xticklabels(x_labels, rotation=10)
ax.set_ylim(0, 1.1)
ax.set_ylabel('F1 Score')
ax.set_title('Per-Class F1 Score - All Models', fontsize=13, fontweight='bold')
ax.axvline(x=len(sub_names) - 0.5 + width * n_models / 2, color='gray', linestyle='dashed')
ax.legend()
sns.despine(ax=ax)
plt.tight_layout()
plt.show()

# Summary table
print('\nSummary Table:')
pivot = f1_df.pivot(index='Subreddit', columns='Model', values='F1').reindex(sub_names)
print(pivot.round(3).to_string())

```



Summary Table:

Model	distilbert/distilbert-base-uncased
Subreddit	
Futurology	0.616
MachineLearning	0.980
artificial	0.635
singularity	0.207
MACRO AVG	0.609

DistilBERT outperforms all TF-IDF models on every class. The biggest improvements are on Futurology and singularity, where shared vocabulary limits TF-IDF the most.

Discussion

DistilBERT significantly outperforms TF-IDF (76% vs ~67% accuracy).

MachineLearning's near-perfect F1 (~0.99) confirms its technical vocabulary is genuinely distinct from the other three. singularity remains the hardest class as it overlaps linguistically with every other community.

All Models Comparison: TF-IDF vs BERT

Re-training the three TF-IDF models from saved artifacts and comparing all models on the same test split for a clean side-by-side view.

```
In [73]: import scipy.sparse as sp
from sklearn.naive_bayes import MultinomialNB
from sklearn.linear_model import LogisticRegression
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.metrics import f1_score
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load TF-IDF train/test artifacts saved in notebook 03
X_train_tfidf = sp.load_npz('data/tfidf_artifacts/X_train.npz')
X_test_tfidf = sp.load_npz('data/tfidf_artifacts/X_test.npz')
y_train_str = np.load('data/tfidf_artifacts/y_train.npy', allow_pickle=True)
y_test_str = np.load('data/tfidf_artifacts/y_test.npy', allow_pickle=True)

SUB_ORDER = ['Futurology', 'MachineLearning', 'artificial', 'singularity']

# — Re-train the three TF-IDF models
nb_model = MultinomialNB(alpha=0.1)
nb_model.fit(X_train_tfidf, y_train_str)

lr_model = LogisticRegression(C=2, class_weight='balanced', max_iter=1000,
                              solver='lbfgs', random_state=42)
lr_model.fit(X_train_tfidf, y_train_str)
```

```

svm_base = LinearSVC(C=5, class_weight='balanced', max_iter=2000, random_st
svm_model = CalibratedClassifierCV(svm_base, cv=3)
svm_model.fit(X_train_tfidf, y_train_str)

# Map BERT integer predictions back to subreddit name strings

bert_preds_named = {
    name: le.inverse_transform(y_pred_int)
    for name, y_pred_int in model_preds_dict.items()
}

# Combine all five models
all_models_preds = {
    'Naive Bayes\n(TF-IDF)': (nb_model.predict(X_test_tfidf), y_test_s
    'Logistic Reg\n(TF-IDF)': (lr_model.predict(X_test_tfidf), y_test_s
    'LinearSVC\n(TF-IDF)': (svm_model.predict(X_test_tfidf), y_test_s
    **{k + '\n(BERT)': (v, y_test_str) for k, v in bert_preds_named.items()}
}

# Compute per-class and macro F1 for every model
f1_rows = []
for model_name, (y_pred, y_true) in all_models_preds.items():
    for sub, f1 in zip(SUB_ORDER, f1_score(y_true, y_pred, labels=SUB_ORDER,
        f1_rows.append({'Model': model_name, 'Subreddit': sub, 'F1': f1})
    f1_rows.append({
        'Model': model_name, 'Subreddit': 'MACRO AVG',
        'F1': f1_score(y_true, y_pred, average='macro')
    })

f1_df = pd.DataFrame(f1_rows)

# Bar chart – all five models side by side
x_labels = SUB_ORDER + ['MACRO AVG']
x = np.arange(len(x_labels))
model_list = list(all_models_preds.keys())
n_models = len(model_list)
width = 0.8 / n_models
palette = ['#6baed6', '#fd8d3c', '#74c476', '#9e9ac8', '#fdae6b']

fig, ax = plt.subplots(figsize=(15, 5))

for i, (model_name, color) in enumerate(zip(model_list, palette)):
    vals = [f1_df[(f1_df.Model == model_name) & (f1_df.Subreddit == s)]['F1'
        for s in x_labels]
    bars = ax.bar(x + i * width, vals, width, label=model_name.replace('\n',
        color=color, edgecolor='white')
    for bar, v in zip(bars, vals):
        ax.text(bar.get_x() + bar.get_width() / 2, bar.get_height() + 0.01,
            f'{v:.2f}', ha='center', va='bottom', fontsize=7)

ax.set_xticks(x + width * (n_models - 1) / 2)
ax.set_xticklabels(x_labels, rotation=10)
ax.set_ylim(0, 1.15)
ax.set_ylabel('F1 Score')
ax.set_title('All Five Models – Per-Class and Macro F1 (TF-IDF vs BERT)',
    fontsize=13, fontweight='bold')

```

```

ax.axvline(x=len(SUB_ORDER) - 0.5 + width * n_models / 2,
           color='gray', linestyle='--', alpha=0.5)
ax.legend(loc='upper left', fontsize=8)
sns.despine(ax=ax)
plt.tight_layout()
plt.show()

# Summary table
print("\nMacro F1 Summary – all models ranked:")
macro_df = (f1_df[f1_df.Subreddit == 'MACRO AVG'][['Model', 'F1']]
            .sort_values('F1', ascending=False)
            .reset_index(drop=True))
macro_df.columns = ['Model', 'Macro F1']
macro_df['Model'] = macro_df['Model'].str.replace('\n', ' ')
print(macro_df.round(3).to_string(index=False))
print("\nRandom baseline (4 equal classes): 0.250 | Majority-class baseline")

```



Macro F1 Summary – all models ranked:

Model	Macro F1
Logistic Reg (TF-IDF)	0.644
LinearSVC (TF-IDF)	0.640
Naive Bayes (TF-IDF)	0.622
distilbert/distilbert-base-uncased (BERT)	0.609

Random baseline (4 equal classes): 0.250 | Majority-class baseline: ~0.336

DistilBERT achieves the best result: a **~76% accuracy**, **Macro F1 ≈ 0.73**, more than double the 25% random baseline and ~9 points above our best TF-IDF model.

The ~75–76% accuracy showcases real overlap in the data. These four subreddits all discuss AI, and some posts are genuinely ambiguous even to a human reader. That's not a model failure but rather it's what the classification performance is telling us.